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Phase-Conditioned Implicit Neural Representations for Three-Dimensional Left Ventricular Reconstruction and Myocardial Wall Thickness Estimation

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Master in Artificial Intelligence

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ISCTE-Instituto Universitário de Lisboa

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TECNOLOGIAS
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Department of Information Science and Technology

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To my grandparents, my parents, and my brother.

For your love, your patience, and the support that made this possible.

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Resumo

A análise da geometria e da espessura da parede do ventrículo esquerdo é relevante para a caracterização da estrutura e função cardíacas. Contudo, a ressonância magnética cardíaca em eixo curto fornece uma representação bidimensional através de cortes espacialmente separados, tornando necessária a reconstrução tridimensional da geometria entre os cortes disponíveis. Embora a interpolação permita estabelecer continuidade entre contornos, não garante a preservação da anatomia ventricular. Além disso, a avaliação clínica da espessura miocárdica baseia-se frequentemente em medições localizadas e na análise visual. Neste contexto, o modelo de 17 segmentos da American Heart Association (AHA) constitui uma referência para a avaliação regional do ventrículo esquerdo.

Nesta dissertação, propõe-se uma abordagem baseada em aprendizagem automática para a reconstrução tridimensional do ventrículo esquerdo a partir de contornos de ressonância magnética cardíaca. Utiliza-se uma representação neuronal implícita para descrever continuamente as superfícies endocárdica e epicárdica nas fases de telediástole e telessístole. O modelo é inicialmente treinado com formas ventriculares sintéticas e posteriormente ajustado com segmentações clínicas. A espessura da parede miocárdica é estimada a partir da geometria reconstruída, permitindo a análise local e regional segundo os 17 segmentos da AHA.

O desempenho da abordagem é avaliado por comparação com métodos baseados em funções de base radial e com um modelo estatístico de forma. Os resultados mostram que as superfícies reconstruídas acompanham os contornos observados, com maior concordância entre métodos na cavidade ventricular do que na parede miocárdica. Verifica-se também que as estimativas da espessura dependem da geometria reconstruída e do método de cálculo, pelo que uma correlação elevada não implica necessariamente concordância entre medições. A análise segmentar permite ainda caracterizar as variações regionais da espessura entre a telediástole e a telessístole.

Este trabalho explora, assim, o potencial das representações neuronais implícitas na reconstrução tridimensional do ventrículo esquerdo e na quantificação da espessura miocárdica. Contudo, na ausência de uma referência anatómica tridimensional independente, não é possível determinar conclusivamente qual dos métodos produz a reconstrução mais próxima da anatomia real. Os resultados evidenciam, ainda assim, a influência das opções de modelação geométrica nas estimativas da espessura e a importância de considerar o método de reconstrução na interpretação das medições.

Palavras-chave: Ressonância magnética cardíaca; ventrículo esquerdo; reconstrução tridimensional; representações neuronais implícitas; funções de distância com sinal; espessura da parede miocárdica.

Abstract

The analysis of left ventricular geometry and myocardial wall thickness is relevant to the characterisation of cardiac structure and function. However, short-axis cardiac magnetic resonance imaging provides a two-dimensional representation through spatially separated slices, making three-dimensional reconstruction necessary to recover the geometry between available slices. Although interpolation methods establish continuity between contours, they do not guarantee the preservation of ventricular anatomy. Furthermore, clinical assessment of myocardial wall thickness often relies on local measurements and visual analysis. In this context, the 17-segment model of the American Heart Association (AHA) provides a reference for regional assessment of the left ventricle.

This dissertation proposes a machine learning approach for three-dimensional left ventricular reconstruction from cardiac magnetic resonance contours. An implicit neural representation is used to continuously describe the endocardial and epicardial surfaces at end-diastole and end-systole. The model is initially trained on synthetic ventricular shapes and subsequently fine-tuned using clinical segmentations. Myocardial wall thickness is estimated from the reconstructed geometry, enabling local and regional analysis according to the 17 AHA segments.

The proposed approach is evaluated against radial basis function methods and a statistical shape model. The results show that the reconstructed surfaces follow the observed contours, with greater agreement between methods in the ventricular cavity than in the myocardial wall. Thickness estimates are also found to depend on both the reconstructed geometry and the calculation method, meaning that high correlation does not necessarily imply agreement between measurements. Segmental analysis further characterises regional variations in wall thickness between end-diastole and end-systole.

This work explores the potential of implicit neural representations for three-dimensional left ventricular reconstruction and myocardial wall thickness quantification. However, in the absence of an independent three-dimensional anatomical reference, it is not possible to conclusively determine which method produces the reconstruction closest to the true anatomy. Nevertheless, the results highlight the influence of geometric modelling choices on thickness estimates and the importance of considering the reconstruction method when interpreting the resulting measurements.

Keywords: Cardiac magnetic resonance imaging; left ventricle; three-dimensional reconstruction; implicit neural representations; signed distance functions; myocardial wall thickness.

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List of Acronyms

Acronym	Definition
ACDC	Automated Cardiac Diagnosis Challenge
AHA	American Heart Association
CI	Confidence Interval
DSC	Dice Similarity Coefficient
ED	End-Diastole
EDT	Euclidean Distance Transform
ES	End-Systole
HCM	Hypertrophic Cardiomyopathy
HD95	95th-percentile Hausdorff Distance
ICC	Intraclass Correlation Coefficient
INR	Implicit Neural Representation
LoA	Limits of Agreement
LV	Left Ventricle
M&Ms-2	Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Segmentation challenge (second edition)
MAE	Mean Absolute Error
MRI	Magnetic Resonance Imaging
NIfTI	Neuroimaging Informatics Technology Initiative
NOR	Normal cohort
PCA	Principal Component Analysis
PDE	Partial Differential Equation
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RAS	Right-Anterior-Superior
RBF	Radial Basis Function
ReLU	Rectified Linear Unit
RMSE	Root-Mean-Square Error
RQ	Research Question
RV	Right Ventricle
SAX	Short-Axis
SD	Standard Deviation
SDF	Signed Distance Function
SSM	Statistical Shape Model

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CHAPTER 1

Introduction

This chapter motivates the reconstruction of continuous left-ventricular geometry from contours delineated on widely spaced cardiac magnetic resonance slices and defines the research problem. It then presents the research questions, the contributions of the thesis, and the organisation of the remaining chapters.

1.1. Motivation

Understanding cardiac structure is fundamentally a geometric problem. The left ventricle (LV), the chamber that pumps oxygenated blood into the body, is a three-dimensional structure whose shape, size, and wall thickness change continuously throughout the cardiac cycle. At end-diastole (ED), the ventricle has filled and is at its largest; at end-systole (ES), it has contracted, so the cavity is smallest and the muscular wall, the myocardium, is at its thickest. The difference between these two states is what quantifies ventricular pumping function: cavity volumes at ED and ES give the stroke volume and ejection fraction, while the volume of the myocardium gives its mass [1]. Beyond these global indices, the geometry of the ventricle carries diagnostic information of its own. Myocardial wall thickness is related to the diastolic behaviour of the ventricle [2], an abnormally thick wall is the defining feature of hypertrophic cardiomyopathy [3], and variations in ventricular shape and thickness have been associated with structural remodelling and cardiovascular risk at the population level [4, 5].

Cardiac magnetic resonance imaging (MRI) is the preferred imaging method for measuring these quantities. It is non-invasive, does not use ionising radiation, and offers good contrast between blood and myocardium, which is why it is regarded as the reference standard for ventricular volumes, mass, and function [1, 6]. In routine cine acquisitions, the LV is imaged in short-axis (SAX) planes perpendicular to its long axis, stacked from the base, the wide end where the valves sit, towards the apex, the pointed tip, and repeated over the cardiac cycle so that both ED and ES are captured.

The images themselves, however, do not yield clinical measurements directly. Volumes, mass, and wall thickness are computed only after the relevant boundaries have been outlined on every slice: the endocardium, the inner boundary of the blood-filled cavity, and the epicardium, the outer boundary of the myocardium. This outlining, called segmentation, is therefore the step on which all later measurements depend. Drawing these outlines manually is time-consuming, and the outlines can differ between experts or when the same expert repeats the task [6]. These differences limit reproducibility, while the time required makes manual segmentation less practical as the number of examinations grows. These limitations have motivated a sustained effort towards automated segmentation, driven by public benchmarks and challenges [1, 7, 8]. The need is most evident in large cohorts such as the UK Biobank, where thousands of subjects must be analysed consistently [9], and where fully convolutional networks

now produce segmentations whose derived measurements approach the agreement observed between human experts [10].

Segmentation, however, answers a two-dimensional question. It produces a set of contours on separate planes, whereas the quantities of clinical interest describe a continuous three-dimensional organ. Regional analysis, such as reporting wall thickness according to the American Heart Association (AHA) 17-segment model [11], and the construction of population atlases that compare anatomy across subjects [5, 12] both require a description of the ventricle that extends beyond the individual slices. How that description is obtained from segmented contours is the problem addressed in this thesis.

1.2. Problem Statement

Reconstructing the LV in three dimensions from segmented SAX contours is a long-standing problem in cardiac image analysis. A cine SAX stack typically resolves the ventricle at 1–2 mm within each slice, but consecutive slices lie 6–10 mm apart [6]. Segmentation therefore delineates the myocardium accurately where it is imaged, yet leaves the geometry between slices, including the curvature of the apex and the closure of the base, unobserved. In this thesis, contours of this kind are called sparse: they describe the ventricle in detail within each slice but only at widely spaced positions along its long axis. The contours alone cannot uniquely determine a continuous surface: many different surfaces can pass through the same contours, and the result depends on the assumptions used to fill the gaps.

Several families of methods supply these assumptions. Contour lofting joins corresponding points on adjacent slices and passes exactly through the observations, but produces piecewise-linear geometry between them [6]. Statistical shape models (SSMs) restrict the surface to the modes of variation of a training population [13, 14, 15], which stabilises the reconstruction when observations are sparse but can draw atypical anatomy towards the population mean. Radial basis function (RBF) interpolation defines the surface as the zero level set of a smooth function fitted to the contour points [16], trading regional detail for smoothness. More recently, implicit neural representations (INRs), which encode a shape as the set of points where a function computed by a neural network is zero [17, 18], have been used to reconstruct cardiac surfaces from images with widely spaced slices and from sparse segmentations [19, 20]. Closed LV surfaces from sparse contours are therefore not new in themselves; the approaches differ in the prior that fills the unobserved geometry, and hence in how far a measurement on that geometry reflects the data rather than the prior.

The difference matters most when the quantity of interest is myocardial wall thickness. Thickness is measured between two surfaces, so the endocardium and epicardium must be reconstructed as a consistent pair whose separation remains positive and spatially coherent; surfaces reconstructed independently may each be plausible and still intersect or collapse the wall between them [19]. Priors are also often tied to a single cardiac phase, and a shape model built from diastolic anatomy cannot describe the contracted ventricle at ES without imposing a diastolic shape space on systolic observations. Moreover, a discrepancy of one or two millimetres between two reconstructions of the same contours is small relative to the cavity but substantial relative to the wall. Surface-distance and overlap metrics, the usual

criteria for reconstruction quality [21], do not by themselves show how such discrepancies propagate into thickness measurements.

Public data impose a further constraint. Benchmarks such as the Automated Cardiac Diagnosis Challenge (ACDC) and the second Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Segmentation challenge (M&Ms-2) provide images and segmentation masks for individual slices [1, 22], that is, label images in which each pixel is marked as cavity, myocardium, or background, rather than complete three-dimensional LV meshes, surfaces made of connected triangles. Without an independent three-dimensional reference, a learned model must be supervised with surfaces fitted to the same contours and assessed only through fidelity to the observed slices and agreement with alternative reconstructions.

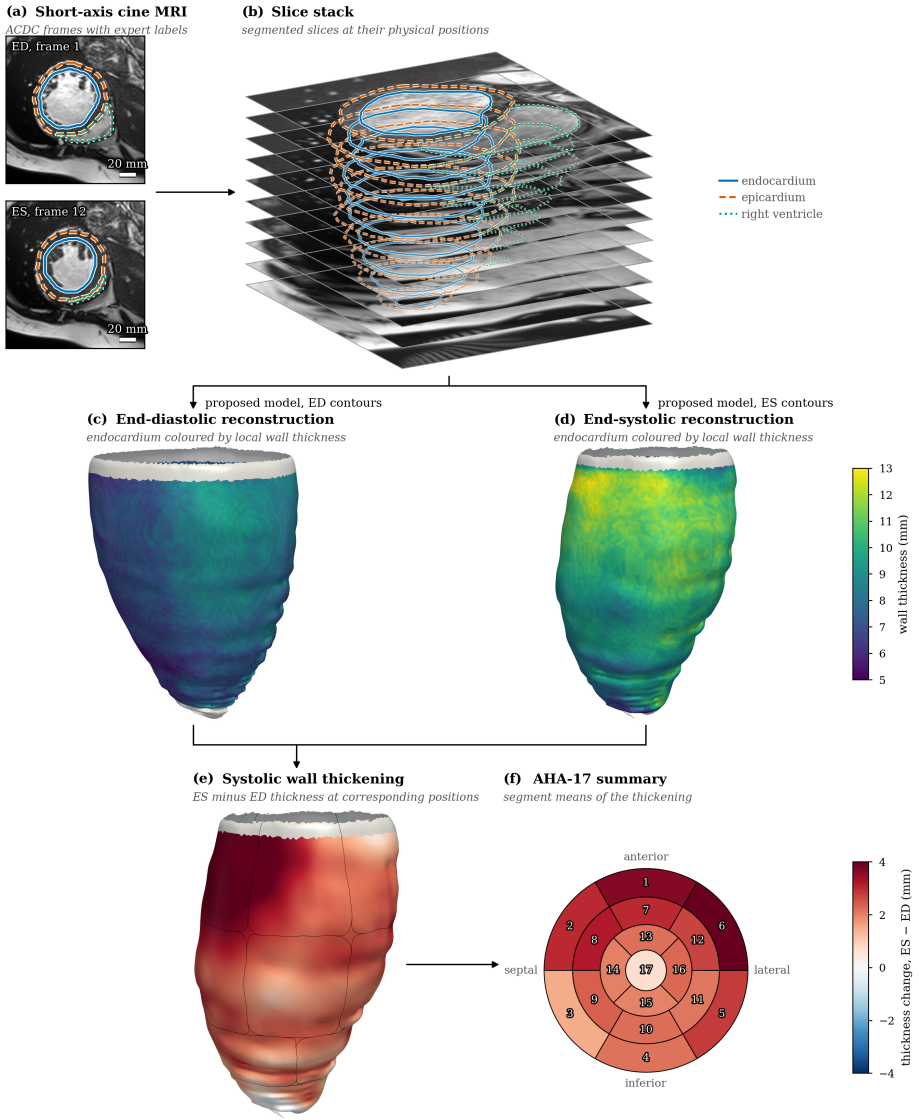


Figure 1.1. Overview of the proposed pipeline. (a) Segmented SAX slices at ED and ES. (b) The slices placed along the LV long axis form a sparse stack of contours, from which the proposed model reconstructs endocardial and epicardial surfaces for (c) ED and (d) ES, coloured by local wall thickness. (e) Wall thickening from ED to ES on the surface, with AHA 17-segment boundaries, and (f) its summary per segment.

This thesis addresses the problem with a phase-conditioned INR, trained on synthetic LV geometries sampled from an SSM and on meshes derived from real cardiac MRI segmentations, as illustrated in Figure 1.1. Rather than introducing a new representation, it adapts the implicit formulation to the constraints described above. The proposed model takes sparse contours as input and represents the endocardial and epicardial surfaces as continuous three-dimensional functions that can be queried anywhere, not only at the acquired slices. Conditioning on cardiac phase lets one model represent both ED and ES, which differ markedly in cavity size and wall thickness, and defining the epicardium through a positive offset from the endocardium keeps the outer surface outside the inner one by construction.

Because no independent reference exists, the evaluation is comparative. The reconstruction is assessed against the observed contours and against RBF and SSM reconstructions of the same contours, and slices are removed from the input to measure how much each reconstruction changes when fewer slices are available (its sensitivity to slice removal). Wall thickness is then measured on each geometry with several methods. This design separates the effect of the reconstruction prior from that of the measurement method, and asks not only whether the proposed model produces continuous surfaces, but whether the resulting geometry preserves quantities relevant to cardiac analysis. The aim is not to replace clinical interpretation, but to investigate whether continuous geometric representations can provide a more consistent basis for quantitative analysis of sparse cardiac MRI.

1.3. Research Questions

As Section 1.2 shows, a closed LV surface can already be obtained from sparse contours by RBF interpolation, SSM fitting, or an implicit neural model. The open questions concern what a learned prior contributes relative to these alternatives, whether it can satisfy the paired-surface and multi-phase requirements of wall-thickness analysis, and how strongly the resulting measurements depend on choices made during reconstruction and measurement. This thesis therefore addresses the following research questions (RQs):

RQ1. How does an implicit reconstruction learned from synthetic and clinical LV geometry compare with RBF and SSM reconstructions of the same sparse SAX contours in fidelity to the observed contours, agreement on the geometry inferred between slices, and sensitivity to the number of available slices?

RQ2. Can a single phase-conditioned model reconstruct paired endocardial and epicardial surfaces at both ED and ES while maintaining a positive wall thickness, without fitting a separate model to each phase or reconstructing the two surfaces independently?

RQ3. To what extent does myocardial wall thickness measured on the reconstructed geometry depend on the reconstruction prior and on the thickness method, and are regional and pathology-related differences in thickness retained?

1.4. Contributions

The thesis makes three contributions: a shared process for preparing training data, an adaptation of implicit reconstruction to paired LV surfaces, and an evaluation approach that distinguishes geometric and measurement consistency.

A shared data pipeline is developed for synthetic LV shapes sampled from an SSM and clinical ED and ES segmentations from ACDC and M&Ms-2. Both sources are converted into input contours and target surfaces in the same format, allowing synthetic pre-training and clinical fine-tuning without changing how the data are represented. The clinical surfaces are fitted to the observed contours and used to train the model; they are not independent three-dimensional anatomical ground truth.

The model adapts established point-set and implicit-representation methods by summarising endocardial and epicardial contour evidence separately and combining a global shape description with local contour information. A cardiac-phase input allows one model to represent both ED and ES, while a positive offset links the endocardial and epicardial fields. This offset is trained directly using the separation between the target surfaces. The contribution is this task-specific combination, not the underlying methods themselves.

An evaluation approach is also established to separate reconstruction from measurement. It examines fidelity to the observed contours, agreement with RBF and SSM reconstructions of the same inputs, and sensitivity to slice removal. Four wall-thickness methods are compared on the same geometry before local measurements are summarised in AHA-17 regions and examined across cardiac phases and patient groups. In the absence of independent three-dimensional ground truth, these comparisons characterise consistency and dependence on methodological choices rather than establish anatomical accuracy or diagnostic performance.

1.5. Thesis Organisation

The remaining chapters follow the progression from the reconstruction problem to its evaluation. Chapter 2 reviews cardiac MRI-based reconstruction and myocardial wall-thickness measurement, providing the context for the methodological choices. Chapter 3 describes the data preparation, the proposed model, its training and inference, and the methods used to measure wall thickness. Chapter 4 evaluates the reconstructed geometry, how much it changes when slices are removed, and the resulting local and regional thickness measurements. Finally, Chapter 5 answers the research questions, summarises the contributions and limitations, and identifies directions for future work.

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CHAPTER 2

Literature Review

This chapter describes the review methodology and examines prior work on reconstructing paired LV surfaces from sparse cardiac MRI observations and measuring the myocardial wall represented by those surfaces.

2.1. Review Methodology

The review was designed to compare methods relevant to sparse cardiac MRI, three-dimensional shape reconstruction, geometric learning, and myocardial wall-thickness measurement, rather than to provide an exhaustive systematic review.

Scopus, PubMed, and IEEE Xplore were searched, with Google Scholar used for backward and forward citation chaining. The search included foundational work published from 2000 onwards and selected studies available up to 2026. Although recent work was prioritised, earlier publications were retained where they establish concepts used by later methods. The resulting review is selective and does not claim exhaustive coverage of the literature published through 2026.

Search terms combined anatomy and imaging concepts (“cardiac MRI”, “left ventricle”, “short-axis”, “SAX”, “myocardium”) with reconstruction terms (“3D reconstruction”, “mesh reconstruction”, “shape model”, “implicit surface”, “signed distance function”), learning terms (“deep learning”, “graph neural network”, “point cloud”, “neural representation”), and measurement terms (“wall thickness”, “myocardial thickness”, “Laplace equation”, “transmural”).

The refined query applied consistently across databases was:
(“cardiac MRI” OR “left ventricle” OR “short-axis”) AND (“3D reconstruction” OR “statistical shape model” OR “signed distance function” OR “graph neural network” OR “implicit representation”) AND (“wall thickness” OR “mesh” OR “deep learning”)

Studies were included when they addressed LV three-dimensional reconstruction or shape modelling, proposed or evaluated myocardial wall-thickness methods, developed geometric learning methods applicable to anatomical meshes, or introduced benchmark cardiac MRI data relevant to LV analysis. Studies focused only on segmentation, on non-cardiac anatomy without transferable methodology, or lacking peer review or sufficient methodological detail were excluded. Publications before 2000 were considered only when they provided foundational methods directly relevant to the reviewed approaches.

The search returned 142 records. After duplicate removal and screening, 28 studies met the selection criteria; backward citation searching added 7, giving a final corpus of 35 publications. Figure 2.1 summarises this process as a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram [23].

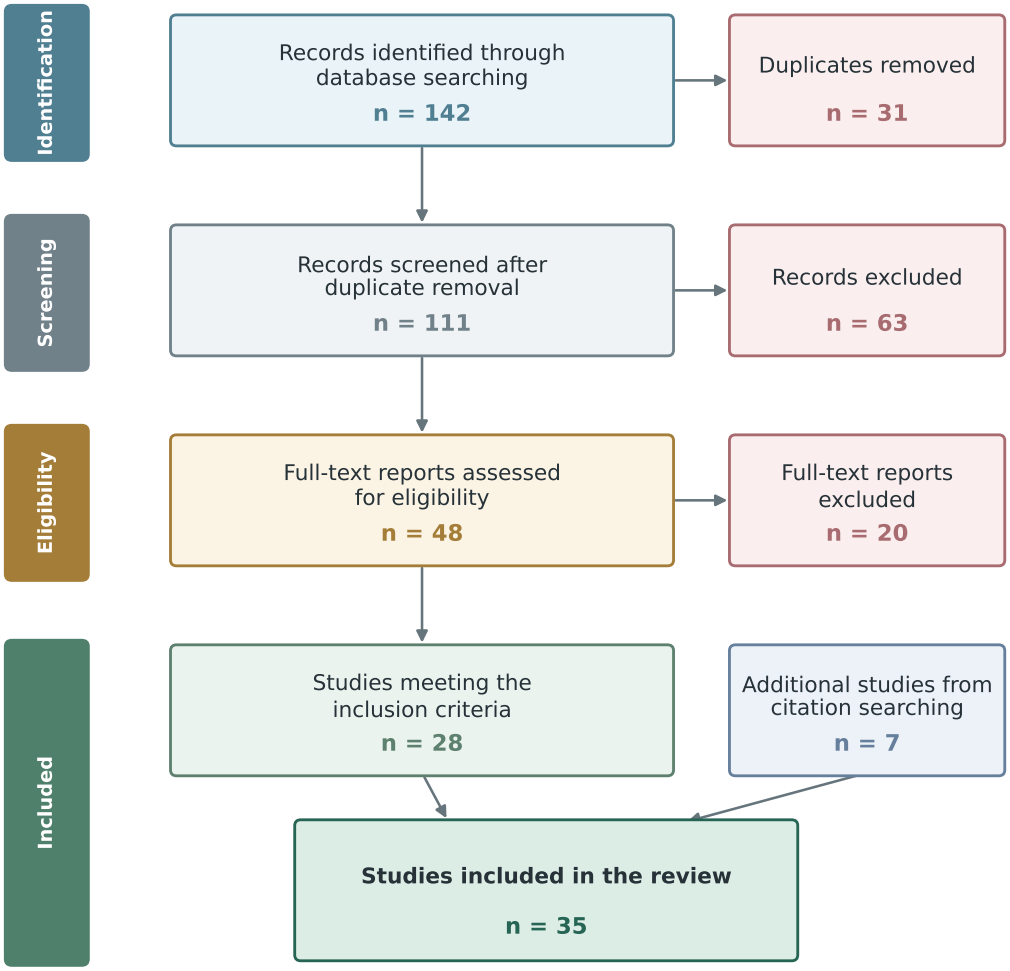


Figure 2.1. PRISMA flow diagram summarising the selection of the 35 publications included in the review.

The selected studies cover cardiac MRI data and segmentation, SSMs, learned 3D reconstruction, and wall-thickness measurement. The next section relates these areas to the specific problem of recovering paired LV surfaces from sparse contours.

2.2. Technical Background

The central distinction in this review is between anatomy observed on acquired MRI slices and geometry that must be inferred between them. This distinction connects the constraints of paired LV surface reconstruction to the available representations of three-dimensional anatomy.

Cine cardiac MRI captures the LV at several points in the cardiac cycle. Of these, ED, when the cavity is largest, and ES, when it is smallest, provide a useful description of ventricular contraction and are the phases labelled in benchmarks such as ACDC [1].

In a SAX acquisition, each plane samples a cross-section perpendicular to the ventricular long axis. The anatomy is resolved in detail within a slice, but the slices are much farther apart than the pixel spacing within each slice, producing an unevenly sampled image stack [6]. Segmentation can therefore recover the myocardium where it is observed without determining how its boundaries curve, taper, or change in the unobserved intervals.

Each slice contributes an endocardial contour around the cavity and an epicardial contour around the outer myocardium. In three dimensions, these must form nested surfaces with a positive separation. A reconstruction that treats the surfaces independently may produce individually plausible boundaries while still creating an intersection, reversal, or local collapse in the wall between them, a difficulty also addressed by recent joint implicit reconstructions of cardiac structures [19].

The contours alone do not determine one unique continuous surface: many smooth surfaces can pass through the same contour rings. Smoothing stabilises the reconstruction but can remove genuine regional variation; a strong population prior can bridge missing regions but may pull abnormal anatomy towards common shape; and close contour fitting can preserve segmentation noise. A useful method must balance these competing constraints rather than optimise any one of them in isolation [6, 15].

The same reasoning determines how reconstruction should be evaluated. The surfaces should agree with observed anatomy, remain closed where volume is needed, preserve subject-specific shape, and form a valid wall at both phases. No single overlap or surface-distance measure assesses all of these properties, so geometric distance, volumetric agreement, topology, and the stability of wall-derived measurements must be considered together [15, 19].

The representation chosen for three-dimensional anatomy determines both what can be measured and how missing geometry is inferred. Voxel volumes are well suited to image-based segmentation and overlap analysis, but retain the spatial limits of the sampling grid. Surface meshes support direct geometric and regional measurements, although remeshing, smoothing, and repair can alter the anatomy being evaluated. Neither representation alone determines the geometry between sparsely acquired slices.

SSMs address this missing information by constraining meshes with population-level modes of variation [13, 15]. They can therefore provide a geometric prior or synthetic training examples, but remain limited by the topology and anatomical variation represented in the source atlas.

Implicit methods offer a different compromise. DeepSDF and Occupancy Networks represent geometry as continuous implicit fields [17, 18], from which meshes can be extracted using methods such as Marching Cubes [24]. Work on Eikonal regularisation and Fourier features further addresses distance-field consistency and the recovery of local spatial variation [25, 26]. By improving field consistency and spatial detail, these developments make implicit representations relevant to sparse-contour reconstruction. The extracted surface nevertheless remains dependent on the learned field and extraction procedure.

Paired LV surfaces introduce an additional constraint: two independently predicted fields do not guarantee a valid ordering. Defining one surface in relation to the other through a positive separation instead represents a reference boundary and a constrained wall. Joint implicit reconstruction of cardiac structures provides one way of modelling related boundaries together, although surface ordering and wall validity must still be assessed explicitly [19].

2.3. Related Work

This section compares how the reviewed methods address the evidence available on acquired slices, the geometry missing between them, and the measurements derived from the reconstructed wall.

2.3.1. Clinical observations and conventional geometric priors

SAX cine MRI is the standard protocol for LV evaluation, producing 2D slices perpendicular to the long axis with high in-plane resolution (1–2 mm) but large inter-slice gaps (6–10 mm) [6]. The resulting anisotropy creates the central reconstruction problem: the observed contours are detailed within each slice, but the geometry between slices, especially near the apex and base, must be inferred. The AHA 17-segment model provides a common anatomical frame for reporting regional measurements [11].

Modern segmentation networks can identify endocardial and epicardial boundaries reliably on the acquired slices. U-Net established the encoder–decoder pattern used by many medical segmentation systems, while nnU-Net showed that much of their configuration can be adapted automatically to a dataset [27, 28].

Although both methods improve the observed contours, they still optimise agreement on sampled image planes. High slice-wise overlap can therefore coexist with uncertain through-plane geometry. Moreover, because the contours become constraints on the reconstructed surface, segmentation errors can propagate into regions where no image evidence is available rather than remaining local to one slice.

ACDC provides expert ED and ES labels across five diagnostic groups, including normal ventricles and hypertrophic cardiomyopathy (HCM) [1]. Its phase and diagnostic variation make it useful for testing whether a reconstruction retains contraction-related and abnormal shape differences. M&Ms-2 focuses on right-ventricular segmentation and extends evaluation across centres, scanner vendors, and diseases [22]. For reconstruction, its multi-source design shows that performance on one acquisition source does not guarantee transfer to another. A plausible surface on one cohort is insufficient evidence of generalisation because acquisition variation affects the contours from which the entire surface is inferred.

At population scale, the UK Biobank provides a large and standardised cardiac MRI resource [9]. Automated segmentation of this cohort has enabled large anatomical studies and statistical atlases [10]. Together, slice-wise clinical observations and population-level atlases provide complementary forms of anatomical evidence.

The two forms are not interchangeable. Clinical labels retain subject-specific and pathological variation, but their through-plane geometry is only indirectly observed. A population atlas supplies complete and consistent surfaces, but it reflects the cohort and processing choices from which it was built. Combining them can reduce the weakness of either source alone, provided that the atlas is used as a prior rather than treated as patient-specific ground truth.

SSMs encode anatomical variability by matching the same anatomical locations across the training population and applying principal component analysis (PCA) to extract the dominant modes of variation [13, 15]. They provide a compact description of plausible shape, but their quality depends on how accurately the same anatomical locations are matched, on alignment, and on the diversity of the population used to construct them.

Cardiac SSMs have progressed from multi-object ventricular models to large population atlases. Matching the same points across subjects allows related structures to be represented consistently [14], while the UK Digital Heart Project atlas captures population variation in ventricular shape and motion [12].

Their modelling of coordinated variation across related cardiac structures is directly relevant to paired-surface reconstruction, where the two boundaries cannot be treated in isolation.

Shape-model studies have also identified variation associated with hypertrophic remodelling, normal anatomical diversity, and myocardial infarction [4, 5, 29]. Their findings show that clinically relevant anatomy is not captured by global volume alone. A population prior must therefore retain regional differences in proportion, apical form, and curvature rather than draw every case towards one mean geometry.

SSMs have two principal roles. They can constrain reconstruction to a learned shape space, or they can generate synthetic geometries by sampling that space [15, 30]. Synthetic sampling provides complete, plausible meshes that can increase the amount of geometric supervision available for learning-based methods.

Synthetic sampling increases the amount of complete geometric supervision, but it cannot introduce anatomy absent from the source atlas. It may also under-represent rare or severe pathology when the retained PCA modes favour common variation. Clinical data remain important for adapting a population prior to subject-specific and pathological anatomy while retaining the surface continuity learned from complete meshes. Combining SSM-derived and clinical data therefore assigns complementary roles to population-level priors and observed anatomical variation.

Before a learned model can be supervised with clinical data, the segmented SAX contours must be related to a three-dimensional surface. Linear contour interpolation, SSM fitting, and RBF fitting offer three different ways to supply the missing through-plane geometry. Their principal difference is the source of the reconstruction prior: linear interpolation uses only neighbouring contours, an SSM uses population-level modes of anatomical variation, and an RBF fit imposes smoothness through a continuous implicit field.

Linear contour lofting first matches points between adjacent rings and then joins them with triangular strips. It is transparent, computationally inexpensive, and passes directly through the observed contours. However, the geometry between two slices is piecewise linear and depends on how the starting points are aligned and which points are matched between rings. Large inter-slice gaps therefore produce abrupt changes in slope, while the apex and base require explicit closure rules. These limitations make lofting useful as a direct interpolation baseline, but less suitable as the sole source of smooth three-dimensional supervision from anisotropic SAX data [6].

SSM fitting instead aligns a template with the observations and optimises its shape coefficients so that the deformed surface approaches the contours. The result retains a fixed topology and population-informed anatomical plausibility, which are valuable when observations are sparse [13, 15]. The same prior also limits the fit: anatomy outside the variation represented by the atlas cannot be recovered faithfully, and fitting requires a model appropriate to the structure and cardiac phase under study. An ED-only SSM cannot provide equivalent supervision for ES without imposing a diastolic shape space on systolic observations.

RBF reconstruction represents the surface as the zero level set of a smooth function fitted to scattered points [16]. Surface samples are assigned zero values, while additional samples displaced to either

side of the contour provide the sign needed to distinguish interior from exterior. The resulting implicit field can be evaluated on a regular grid. Marching Cubes then extracts a mesh from the sampled field [24].

Unlike SSM fitting, this procedure does not require a phase-specific template or fixed mesh correspondence. Unlike linear lofting, it defines the intervening geometry as a smooth field rather than as strips between paired vertices. Its limitations are different: the reconstruction depends on the kernel, smoothing, and off-surface constraints, and strong regularisation can suppress genuine local variation.

2.3.2. Deep learning models for 3D shape reconstruction

Classical reconstruction methods depend on the sampled contours and on the assumptions imposed between slices, particularly near the apex and base, as discussed in Section 2.3.1. Learned reconstruction methods address the same trade-off between observation fidelity and geometric prior through two main families: template deformation and implicit field prediction.

Template methods learn to displace the vertices of a fixed mesh. Pixel2Mesh and Pose2Mesh demonstrate that graph-based deformation can recover dense geometry from sparse image or landmark evidence [31, 32]. In cardiac applications, fixed correspondence is valuable for population analysis and regional measurement [7, 12, 33]. The trade-off is that the output remains tied to the topology and resolution of the chosen template. The same constraint that simplifies correspondence can therefore limit how far the reconstruction departs from the anatomy encoded by that template.

INRs remove the fixed output template by learning a continuous field from which a surface can be extracted at a chosen resolution. DeepSDF and Occupancy Networks established this approach using signed-distance and occupancy formulations, respectively [17, 18]. For sparse cardiac reconstruction, removing the predefined mesh avoids restricting the output geometry to fixed resolution and connectivity. Output flexibility does not, however, guarantee that two predicted LV surfaces remain correctly ordered; anatomical validity must be addressed separately.

Recent cardiac work shows why implicit representations are useful when MRI slices are far apart. In a SAX scan, the detailed shape of the ventricle is observed within each slice, but less information is available between slices. Brás et al. [19] addressed this anisotropy problem with a neural implicit representation that reconstructs the LV and the right ventricle (RV), including the endocardial and epicardial boundaries of the LV myocardium. The study is closely related to sparse-contour reconstruction because both approaches seek continuous cardiac shape from incomplete observations. Brás et al. use image data directly, whereas contour-based approaches operate on segmented point observations.

Muffoletto et al. [20] also reconstruct three-dimensional cardiac shape from sparse segmentations with a neural implicit representation. Their method uses a learned biventricular shape prior and includes both SAX and long-axis views. It is therefore broader than LV-focused signed distance function (SDF) reconstruction. At the time of writing, the authors provide a public project repository, although a complete code and dataset release could not be verified.

The INR literature also provides mechanisms relevant to thin cardiac anatomy. Fourier features improve the recovery of local spatial variation, while Eikonal regularisation promotes a consistent distance

field [25, 26]. In sparse-slice settings, both mechanisms can help mitigate the loss of local surface detail associated with interpolation across large gaps.

Recent cardiac INRs also model motion across the cardiac and respiratory cycles [34, 35, 36, 37]. Their main goal is temporally coherent image or volume reconstruction, whereas other implicit approaches focus specifically on continuous anatomical surface reconstruction conditioned on cardiac phase.

PointNet provides a complementary way to encode unordered observations [38]. Its predictions do not change when the input points are listed in a different order, which is useful for SAX contours because their point ordering is arbitrary.

2.3.3. Wall-thickness measurement techniques

Myocardial wall thickness is a primary clinical biomarker for cardiac remodelling. Grossman et al. [2] demonstrated that wall thickness is a principal determinant of left-ventricular diastolic stiffness, with hypertrophied ventricles exhibiting markedly elevated stiffness proportional to ED wall thickness. More recently, Huelnhagen et al. [39] showed at 7.0T MRI that wall thickness correlates with tissue transverse relaxation properties, providing a non-invasive way to assess myocardial tissue properties. The AHA 17-segment model [11] provides the standard framework for reporting regional thickness values.

The accuracy of wall-thickness measurement depends critically on the quality of the underlying surface reconstruction, as any smoothing or geometric artefact propagates directly into the thickness estimate. The literature describes five families of measurement methods, each with distinct trade-offs.

The simplest approach uses nearest-neighbour distance: for each endocardial vertex, a k-d tree [40] identifies the closest epicardial point, yielding a measurement in $O(N \log M)$ time. Its runtime depends on mesh size and the implementation. Nearest-neighbour distance can provide a lower bound on the true thickness across the wall, also called transmural thickness, because it ignores the anatomical direction of measurement — the nearest point may lie tangentially rather than across the wall.

Ray-based methods address this limitation by casting a ray from each endocardial vertex along its surface normal, the direction perpendicular to the surface, until it intersects the epicardium. The intersection is computed via the Möller–Trumbore algorithm [41], accelerated with a bounding volume hierarchy to $O(\log F)$ per query. Measuring along the surface normal better approximates the direction across the wall but remains sensitive to errors in the normal: a slight rotation can make the measured distance too long or too short. The shape diameter function variant [42] reduces this sensitivity by casting a cone of K rays (typically $K = 7$) at a half-angle of 30° around the normal and reporting the median intersection distance, effectively filtering outlier hits caused by local mesh irregularities.

Volumetric methods operate on voxel grids. The Euclidean Distance Transform (EDT) computes, for each myocardial voxel, the distance to the nearest boundary, and the sum of the endocardial and epicardial distance fields approximates local thickness ($t = D_{\text{endo}} + D_{\text{epi}}$). The estimate is exact for planar parallel boundaries but can be biased in curved regions because the nearest-boundary directions do not necessarily follow a common path across the wall. The method is computationally efficient ($O(V)$ via the Saito–Toriwaki algorithm [43]) but limited by voxel resolution.

Partial differential equation (PDE)-based methods are widely used to estimate thickness across tissue layers. In a method developed for cortical rather than cardiac anatomy, Jones et al. [44] solved Laplace's equation between the two tissue boundaries and used the resulting non-intersecting field lines to establish corresponding paths. The same geometric principle can be transferred to myocardial boundaries, with thickness obtained from the path length between the boundaries along these streamlines. Yezzi and Prince [45] developed a PDE method that calculates distance fields spreading from both boundaries along the Laplacian field. Adding these distances gives the local tissue thickness.

Finally, mesh-based methods that match points between the surfaces and smooth the resulting measurements combine an initial geometric estimate (normal projection or nearest neighbour) with iterative Laplacian smoothing on the mesh connectivity graph. At each iteration, each vertex's thickness is updated as a weighted average of its neighbours, anchored to the initial estimate. Regularisation produces spatially smooth thickness maps suitable for bullseye visualisation, but too much smoothing can reduce genuine local differences in thickness.

Wall-thickness values depend on both the measurement method and the reconstructed surfaces. Smoothing may make a measurement more stable, but it can also change the distance between the endocardial and epicardial surfaces. Similarly, an irregular surface can create thickness variation that does not reflect the myocardium. For this reason, thickness methods should be compared on the same surfaces. Reconstructions should also be compared using the same thickness method. The present work evaluates both reconstruction consistency and agreement between thickness methods.

2.3.4. Summary and research gap

The reviewed literature shows that each component of the LV analysis pipeline is well-studied in isolation: deep learning gives accurate slice-wise segmentation, SSMs encode population-level shape priors, implicit representations provide continuous surface modelling, and multiple wall-thickness algorithms exist. However, limited work integrates these components into a single pipeline for sparse SAX contour-based reconstruction.

A gap remains in methods that jointly address sparse SAX contour encoding, continuous paired endocardial and epicardial surface reconstruction, cardiac-phase conditioning, preservation of positive wall separation, and training with both clinical observations and SSM-derived synthetic geometries. Such an integrated framework would combine the complementary strengths of population-based shape priors, continuous implicit representations, and clinical contour data while supporting wall-thickness measurement beyond the acquired slices.

There is also an evaluation gap. When no complete three-dimensional reference is available, contour agreement only tests the acquired slices. It cannot validate the reconstructed shape between slices. Evaluation should therefore distinguish contour fidelity from agreement between reconstruction methods. Thickness values must also be interpreted as measurements on an inferred wall, not as direct observations. These principles guide the evaluation in the following chapters.

CHAPTER 3

Methodology

This chapter presents the methodology for reconstructing three-dimensional LV geometry from sparse SAX contours and measuring myocardial wall thickness on the resulting surfaces. The approach combines synthetic geometric supervision with clinical MRI-derived data, a phase-conditioned implicit representation, and a comparison of local wall-thickness methods.

3.1. Overview

Each reconstruction starts from SAX cardiac MRI slices on which the LV has been segmented. The annotations trace two boundaries on each slice: the endocardium around the blood-filled cavity and the epicardium around the muscular wall, or myocardium. Together, these rings capture the ventricle at ED, when it is fullest, and ES, when it is most contracted. But the rings are only cross-sections, typically from roughly ten planes separated by several millimetres. The anatomy between them is not observed. The task is therefore to infer a coherent three-dimensional pair of surfaces from sparse contours, then use that geometry to measure wall thickness throughout the ventricle.

Training examples with both sparse contours and independent three-dimensional LV surfaces are not available in the public datasets. Supervision is therefore built from two complementary sources. An SSM, which compactly represents variation in ventricular shape, supplies complete synthetic surfaces for pre-training the model to recover general LV geometry. Clinical contours from ACDC and M&Ms-2 provide patient-specific anatomy in both phases; smooth surfaces fitted to those contours serve as targets for fine-tuning. In this sequence, synthetic shapes teach the model a broad geometric prior, and clinical cases adapt it to observed patient anatomy.

At inference, only the segmented contours are supplied. The model must recover both boundaries in the unobserved regions while keeping the epicardium outside the endocardium. To enforce this relationship throughout the volume, rather than only at the observed slices, the endocardium is represented by an SDF, whose value gives the signed distance to the surface. The epicardial field is defined by subtracting a strictly positive offset from the endocardial field, preventing the surfaces from crossing. A cardiac-phase input lets the same model reconstruct both ED and ES.

Even after reconstruction, thickness requires a measurement choice: on a curved wall, different ways of pairing inner and outer surface points can yield different values. Four established methods are therefore applied to the same geometry and compared for consistency; the selected method is used to summarise thickness in the AHA-17 regions. Since the clinical datasets do not provide independent three-dimensional ground truth, evaluation also compares the reconstructions with the observed contours and with surfaces fitted from those contours. The model components and their adaptation to this task are detailed in Section 3.3 and summarised in Table 3.3.

Figure 3.1 summarises the whole pipeline. Panels (a) and (b) show the two training stages, synthetic pre-training and clinical fine-tuning. Panels (d)–(g) follow a clinical case from its segmented slices and contour stacks to the reconstructed surfaces and local thickness values at ED and ES, and finally combine the two phases into systolic thickening, both on the surface and summarised in the AHA 17-segment layout.

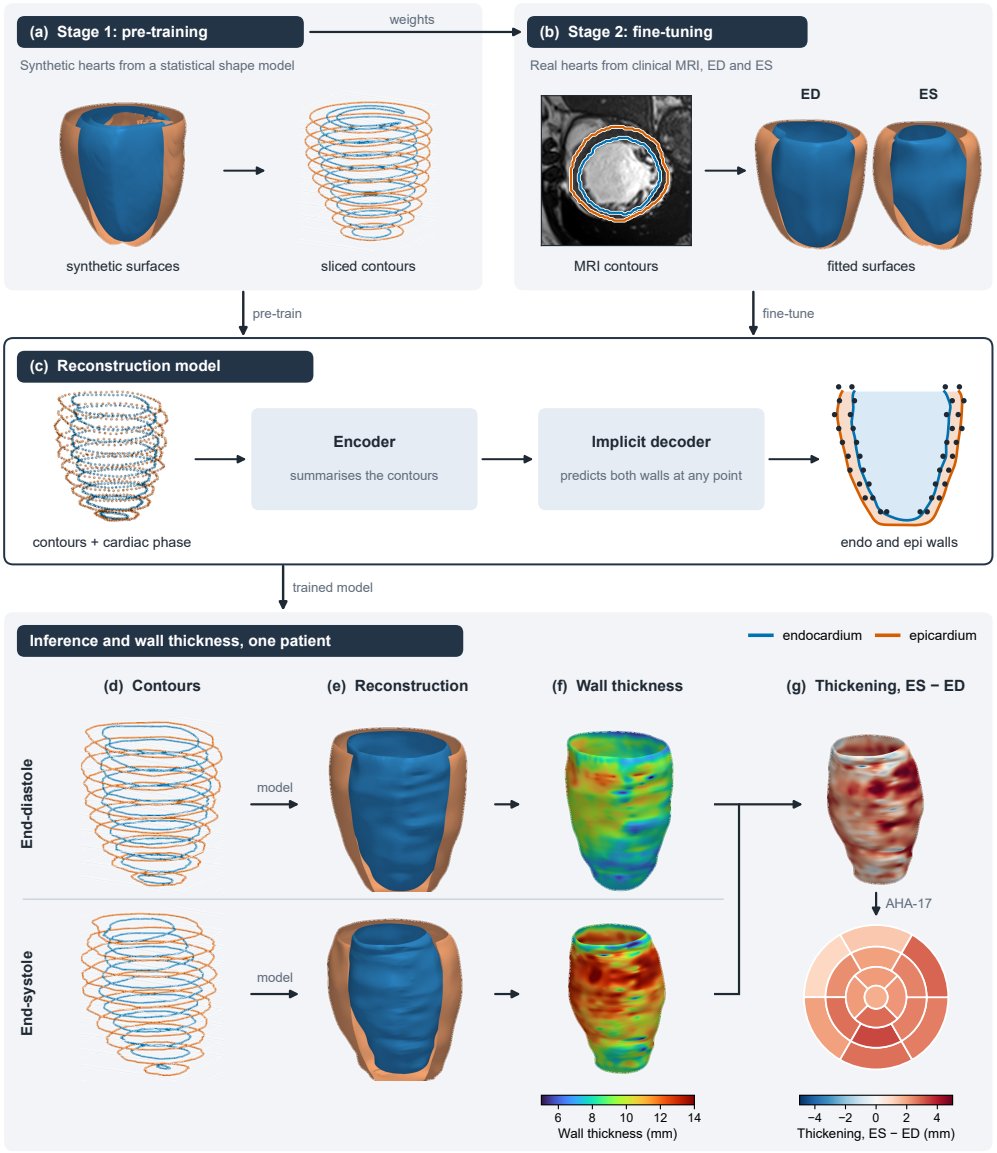


Figure 3.1. Overview of the methodology. (a) Pre-training on synthetic ED surfaces and their SAX contours. (b) Fine-tuning on clinical ED and ES contours with fitted target surfaces. (c) Phase-conditioned reconstruction of the endocardial and epicardial surfaces. (d)–(f) Clinical inference from observed contours to reconstructed surfaces and local Laplace wall thickness at ED and ES, which share one colour scale. (g) Systolic thickening (ES minus ED) on the ES endocardium and the corresponding difference in AHA-17 segment means. Surfaces are smoothed for display only.

The sections that follow develop each stage in turn. Section 3.2 describes how the two training-data sources are constructed and brought into a shared sample format; Section 3.3 and Section 3.4 explain the model, its training, and surface extraction. Section 3.5 then describes the measurement methods and how results are grouped into AHA-17 regions. In Chapter 4, these methods are evaluated against

the observed contours and fitted surfaces, under slice removal, and across cardiac phases and patient groups. Together, the results address the research questions in Section 1.3.

3.2. Data Preparation

Because the model maps contours to surfaces, every training case must contain both: a sparse contour stack as input and a pair of closed three-dimensional surfaces as target. Public cardiac MRI datasets provide only the first half. ACDC and M&Ms-2 supply images and slice-wise segmentation masks, label images in which every pixel is marked as cavity, myocardium, or background, but not the closed ("watertight") endocardial and epicardial surfaces that a signed-distance decoder needs as supervision. Training data were therefore constructed from two sources that fill this gap in complementary ways. The UK Digital Heart Project LV SSM[1] supplies complete surfaces over a controlled range of healthy shapes, from which 1300 synthetic ED cases were generated for pre-training. The clinical segmentations supply patient-specific anatomy in both cardiac phases, and were converted into closed surfaces by fitting smooth implicit functions to their contours, for fine-tuning. The two sources are described in turn, followed by the shared format, augmentation, and division into training, validation, and test sets by patient.

3.2.1. Synthetic ED geometry from the statistical shape model

An SSM describes a population of shapes by a mean shape and a small number of principal modes of variation, the main independent ways in which the shapes of the population differ from that mean [15]. Sampling it yields complete, closed surfaces whose variation is known and controlled, which makes it a convenient source of supervision for pre-training. What it cannot provide matters just as much: the sampled shapes carry no scanner-dependent acquisition effects, no variation in how a clinician would have drawn the contours, and, for the model used here, no systolic geometry. Synthetic pre-training therefore teaches the model what an LV looks like in general, before the clinical stream exposes it to the conditions of the evaluation data.

The synthetic stream draws on the UK Digital Heart Project LV SSM of Bai et al. [12], built from more than 1,000 high-resolution cardiac MRI scans of healthy subjects. It provides a mean endocardial surface of 22 043 vertices (the corner points of the triangles that make up the mesh) together with 100 principal modes of variation, stored as a $66\,129 \times 100$ matrix of mode vectors, and a corresponding vector of 100 eigenvalues λ_i, each measuring how much of the population's variation its mode explains. A new endocardial mesh is generated by

$$X(b) = \bar{X} + \Phi b, \tag{3.1}$$

where $\bar{X}$ is the mean shape flattened to a vector, Φ is the matrix of mode vectors, and $b = [b_1, \ldots, b_{100}]^\top$ is the sampled coefficient vector. The standard deviation (SD) $\sigma_i = \sqrt{\lambda_i}$ sets the scale of variation along mode i. The first modes capture the largest sources of variation in the population: mode 1 mainly controls overall LV size, mode 2 the elongation along the long axis, and higher modes increasingly local deformations (Figure 3.2).

[1] https://github.com/UK-Digital-Heart-Project/Statistical-Shape-Model

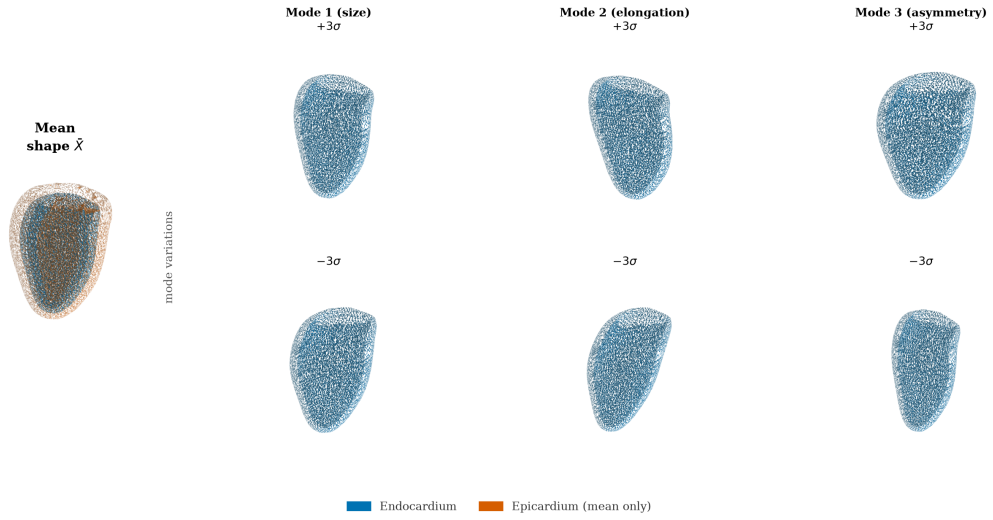


Figure 3.2. UK Digital Heart Project LV SSM mean shape and first three PCA modes at $\pm 3\sigma$.

Of the 100 modes, the first 32 capture 99% of the shape variance and define the sampling space. Each candidate coefficient vector b was drawn by sampling each active component as $b_i \sim \mathcal{N}(0, \sigma_i^2)$ and was accepted only if it satisfied

$$\left|\frac{b_i}{\sigma_i}\right| \leq 3, \qquad \sum_i \left(\frac{b_i}{\sigma_i}\right)^2 \leq \chi^2_{0.99,d}, \tag{3.2}$$

where d is the number of active modes and $\chi^2_{0.99,d}$ is the 99th percentile of the chi-squared distribution with d degrees of freedom. The first condition keeps every mode within three SDs of the mean. The second limits the overall Mahalanobis distance, which measures the combined deviation after each mode has been scaled by its population variability. The threshold retains combinations within the 99th percentile of the model distribution.

Each statistically valid vector is substituted into Equation (3.1). The resulting vertices retain the connectivity of the mean mesh, producing a closed and consistently oriented surface. The shape checks then use the enclosed volume, surface area, sphericity, and the direction of the surface normals at the apex and base. Samples outside limits set from the population data are discarded. Exact volume and area calculations on every fortieth sample confirmed the reliability of the faster descriptors. Of 2048 candidates, 1300 (63%) passed both the statistical and geometric checks.

The SSM provides only the endocardial surface. An epicardial surface was synthesised by moving each endocardial vertex p_i outward along its normal n_i by a distance d_i,

$$q_i = p_i + d_i n_i, \tag{3.3}$$

with d_i set to approximately 10 mm near the base and 5 mm near the apex, plus Gaussian noise with an SD of 0.5 mm so that the wall is not perfectly uniform. This synthetic epicardium exists only to give the decoder a consistent outer surface during pre-training, so that the model learns the general relation between an inner and an outer LV surface before it sees clinical data. It does not reproduce clinical wall thickness and is never used as a reference for any thickness result.

Finally, each surface pair was converted into the form the model will see. Slicing both surfaces at 10 evenly spaced levels along the long axis produces sparse SAX contour rings that mimic a clinical acquisition. For each case, 2048 query points, locations in space at which the decoder will be asked for a value during training, were sampled near the surfaces and in the surrounding volume, and the signed distance to each surface, or an inside/outside label, was recorded at every one of them; these points provide the dense volumetric supervision for the decoder. Figure 3.3 summarises the whole procedure, from the accepted region of coefficient space to the contour rings and query points of one case. The synthetic stream fixes the general shape of the problem; the clinical stream, described next, supplies the anatomy on which the model is evaluated.

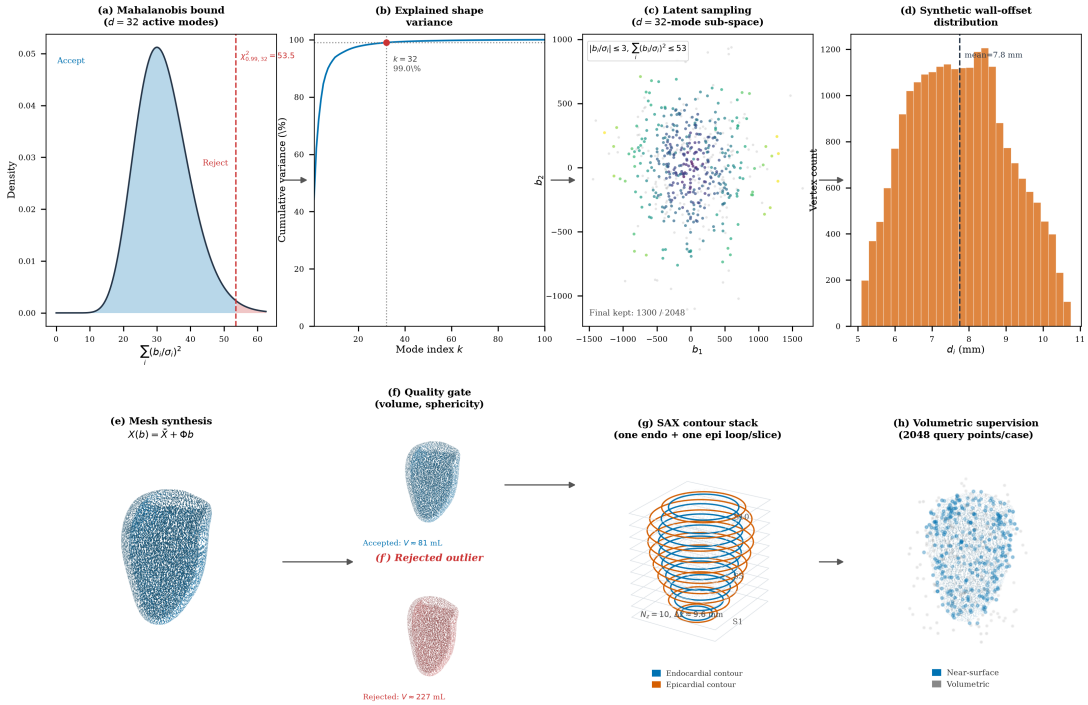


Figure 3.3. Synthetic ED dataset generation, from SSM sampling to geometric supervision. Top row, statistical constraints: (a) distribution of the overall-deviation (Mahalanobis) statistic of Equation (3.2) with the $\chi^2_{0.99,32}$ acceptance threshold; (b) cumulative shape variance, with 32 active modes reaching 99%; (c) accepted coefficient samples in the first two modes; (d) distribution of the synthetic epicardial offset of Equation (3.3). Bottom row, geometric construction: (e) mesh synthesis via Equation (3.1); (f) an accepted shape next to a rejected outlier; (g) the 10 SAX contour levels; (h) the 2048 near-surface and volumetric query points used for decoder supervision.

3.2.2. Clinical ED and ES surfaces from SAX labels

The clinical data come from two public collections, ACDC [1] and M&Ms-2 [22], which together provide 460 patients (100 from ACDC and 360 from M&Ms-2) with outlines of the cavity, the myocardium, and the RV drawn by experts at ED and ES. Both are available for research at no cost after registration and acceptance of a data-use agreement, and the shape model of the previous section is likewise freely available. These data contribute what the synthetic stream lacks: scanner and site variability, the way clinicians actually delineate the wall, patient-specific and in part pathological anatomy, and the contracted geometry of ES.

Segmentation masks are not yet surfaces. Figure 3.4 shows representative ACDC slices with the cavity, myocardium, and right-ventricular labels, and the ED labels of one patient stacked at their true through-plane positions. The labels are defined one slice at a time, and the slices are often several millimetres apart, so stacking them gives a set of sparse rings with nothing in between. Turning these rings into closed surfaces is a reconstruction problem in itself, and the method chosen for it determines what the model is trained to reproduce.

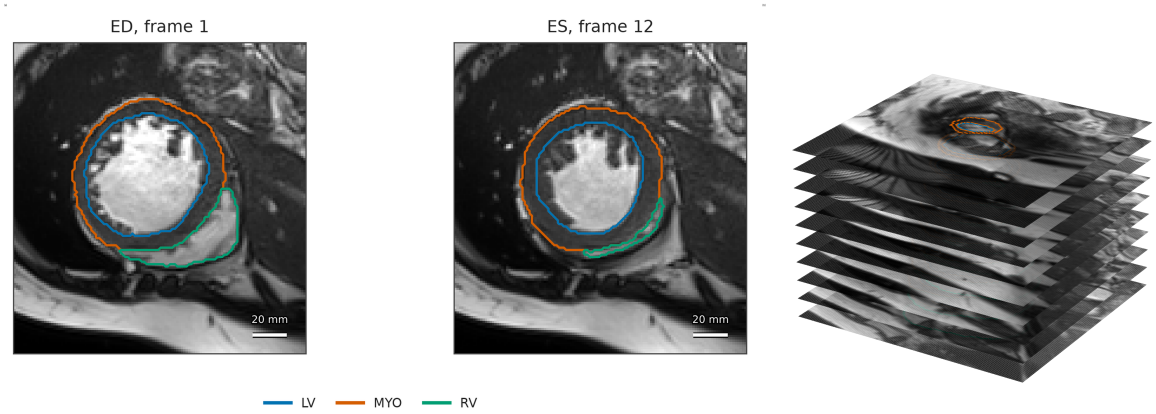


Figure 3.4. Clinical-input view used in the real-data pipeline: (a) representative ACDC ED/ES SAX slices with labels and (b) the corresponding sparse ED SAX stack in physical three-dimensional spacing.

Three options were available. Connecting corresponding points on consecutive rings (lofting) requires a one-to-one pairing of vertices between rings that the data do not provide, and gives a surface that is only piecewise linear between slices. Fitting the SSM would impose a population prior on every target and, because the available model has ED modes only, would either force systolic hearts into a diastolic shape space or treat the two phases differently. Fitting a smooth implicit function to the rings has neither drawback: it needs no vertex pairing and no atlas, and the same procedure applies to ED and ES alike. The clinical targets were therefore built by RBF fitting, which leaves the SSM in its role as the source of synthetic pre-training geometry.

Each segmentation is distributed as a Neuroimaging Informatics Technology Initiative (NIfTI) file, a standard container for medical image volumes. The file stores the label volume together with the physical size and position of its voxels. Each volume is reoriented to the RAS+ anatomical convention, whose axes point towards the patient's right, anterior, and superior sides. Physical voxel spacing is preserved so that all cases remain expressed in millimetres within a common coordinate frame. Because ACDC and M&Ms-2 assign different integer labels to the anatomical structures, the cavity and myocardium labels are mapped to common definitions.

On each acquired SAX slice, the cavity boundary defines the endocardial ring, while the outer boundary of the combined cavity and myocardium defines the epicardial ring. If a label contains several closed boundaries, as can occur with a small detached fragment, only the largest is retained. Each ring is resampled to 60 evenly spaced points and stored at its measured position along the long axis. The resulting points are both the sparse observation supplied to the encoder and the constraints used to construct the target surfaces.

The endocardial and epicardial surfaces are fitted separately. For either surface s, let p_i be a contour point and n_i its outward in-plane normal, estimated from the local contour tangent and pointing away from the ring centre. Following Carr et al. [16], a smooth function f_s is defined over the whole volume as a sum of RBFs, smooth bumps whose value depends only on the distance to a centre c_j, plus a linear term, and the surface is taken as its zero level set, the set of points where the function is zero. With the thin-plate-spline kernel ϕ used here, the function is

$$f_s(x) = \sum_{j=1}^{K} w_j \phi(\|x - c_j\|_2) + a_0 + a^\top x, \qquad \phi(r) = r^2 \log r, \tag{3.4}$$

In Equation (3.4), the centres c_j are the constraint points, and the weights w_j, a_0, and a are chosen so that the function satisfies the constraints. The contour points fix where the function is zero, while paired points placed a distance $d = 2.5$ mm inside and outside each contour point fix its sign and local scale through

$$f_s(p_i) = 0, \qquad f_s(p_i + dn_i) = d, \qquad f_s(p_i - dn_i) = -d. \tag{3.5}$$

The constraints in Equation (3.5) are applied with a regularisation parameter of 0.5. This lets the function pass close to, rather than exactly through, the constraints, reducing its sensitivity to small irregularities in the drawn contours.

Each fitted function is evaluated on a 1.0 mm grid that extends 8.0 mm beyond the rings in-plane and 1.5 mm beyond the first and last ring along the long axis, so that the surface can close just beyond the observed extent. The region where $f_s \leq 0$ is restricted to its valid connected component, discarding small disconnected pockets, and converted to a signed-distance volume. A Gaussian filter with a width of 1.2 mm smooths this volume before marching cubes extracts the zero-level surface [24]. Applying the smoothing before mesh extraction removes small voxel-induced ripples without subsequently moving the mesh vertices.

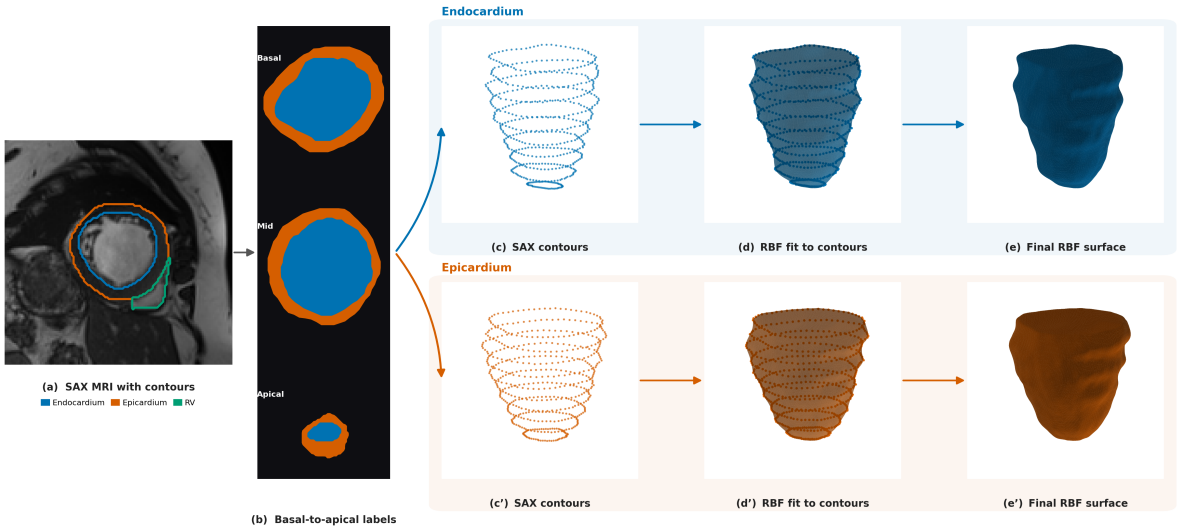


Figure 3.5. Clinical RBF contour-to-surface pipeline. Panel (a) shows one acquired SAX slice with the expert endocardial, epicardial, and right-ventricular contours. Panel (b) shows the corresponding basal, mid-cavity, and apical labels, while panels (c) and (c′) stack the extracted contour rings. Panels (d) and (d′) overlay those constraints on the zero-level surfaces extracted from the smoothed RBF fields, and panels (e) and (e′) show the final topology-checked surfaces used to construct the clinical supervision.

The extracted mesh is accepted unchanged when it forms a single closed shell with consistently oriented faces and no handles or tunnels. Otherwise, only the failed property is repaired. A case is excluded if either surface has fewer than two valid rings, the RBF fit fails, a single closed shell cannot be recovered, the dimensions are implausible, or the epicardium does not fully enclose the endocardium. The complete operations and parameters are listed in Table A.2 of Chapter A, while Figure 3.5 illustrates the separate fitting and validation of the two surfaces.

Both meshes are centred and scaled using the same translation and scale factor. As for the synthetic cases, 2048 query points are sampled near the surfaces and throughout a surrounding box, and their signed distances to both surfaces are recorded.

These fitted surfaces are training targets, not independent anatomical references. Their shape between slices depends on the RBF kernel, off-surface constraints, regularisation, and extraction resolution. Agreement with them therefore measures reproduction of the fitted geometry, not recovery of the true anatomy. Evaluation in Chapter 4 also measures agreement with the observed contours.

3.2.3. Shared sample format, augmentation, and splits

Both streams enter training through the same sample format, the same augmentation rule, and partitions defined at the level of patients.

Table 3.1. Fields of one stored sample, shared by the synthetic and clinical streams, and their role in training.

Field	Origin	Role
Contour points	Synthetic mesh slicing or clinical slice labels	Sparse SAX observation for the encoder
Tissue label	Contour extraction	Distinguishes endocardial from epicardial points during tissue-specific pooling
Phase label	Dataset metadata	Conditions the encoder on ED or ES geometry
Surface samples	Prepared endocardial and epicardial meshes	Surface-fitting and wall-offset supervision
Query points and targets	Near-surface and volumetric mesh sampling	Dense signed-distance and off-surface supervision
Mesh metadata	Preprocessing pipeline	Coordinates, scale, faces, and orientation for evaluation and visualisation

Synthetic and clinical cases use the format in Table 3.1, with 10 SAX levels, 60 points per available ring, and 2048 query points per case. Only the contour coordinates and tissue and phase labels are supplied to the encoder. Surface samples and distance targets are used for training, not as reconstruction inputs. The architecture and loss remain unchanged between pre-training and fine-tuning.

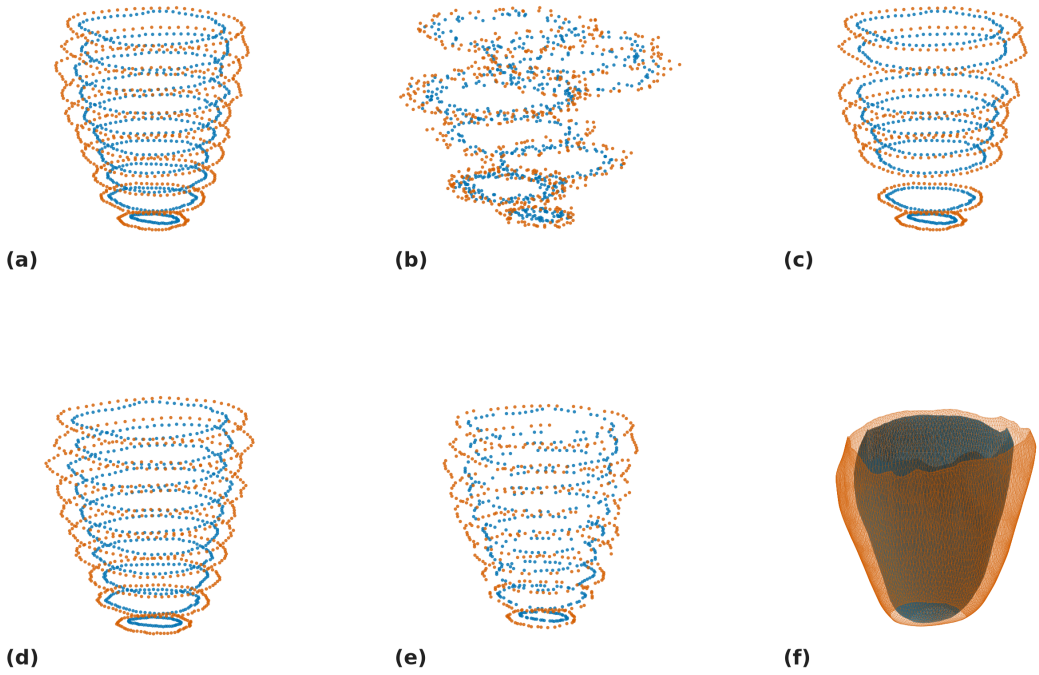


Figure 3.6. Contour-input augmentation: (a) original stack; (b) per-slice translation and point jitter; (c) slice dropout; (d) rotation and scaling; (e) point dropout; (f) fixed target mesh. Endocardium in blue, epicardium in orange.

During training, input contours are modified to represent variation in acquisition and annotation [46, 47]. The modifications include small independent in-plane slice shifts, small random point displacements, global scaling, and point removal. Endocardial and epicardial rings from the same slice are shifted together. Less frequently, whole slices are removed while retaining at least three, or the stack is rotated about its long axis (Figure 3.6). Target meshes, surface samples, query points, and tissue and phase labels remain unchanged. Augmentation is applied to both data sources after the patient-level split, preventing modified test cases from entering training.

Not every clinical patient yields a usable target. A study may contain only long-axis views or too few slices, or its segmentation may fail the plausibility checks of Section 3.2.2. After these checks, 444 of the 460 patients provide a valid ED mesh and 423 a valid ES mesh; together with the 1300 accepted synthetic meshes they form the training data summarised in Table 3.2.

Table 3.2. Number of valid samples per data source and cardiac phase.

		Valid samples		
Source	Type	ED	ES	Role
ACDC and M&Ms-2	Clinical MRI	444	423	Fine-tuning and evaluation
UK Digital Heart SSM	Synthetic meshes	1300	—	Pre-training only

The clinical patients are divided into training, validation, and test sets in a 70 %/15 %/15 % proportion, rounded to whole patients and applied to patients rather than to meshes, so that the ED and ES meshes of one patient always fall in the same set and no information passes from the test patients into training. The

Table 3.3. Model components, the prior work each one builds on, and what is specific to this work.

Component	Builds on	Specific to this work
Input representation	Unordered point-set input [38]	Every point carries a tissue label and the cardiac phase, so one model serves ED and ES
Global encoder	Shared per-point network with max-pooling [38]	Endocardial and epicardial points are pooled separately before being merged into the latent code
Local feature volume	Convolutional feature grid read at the query point [48]	Built from a few hundred contour points averaged into a coarse 16^3 grid; fed to the decoder trunk and to the offset head
Implicit decoder	Latent-conditioned signed-distance network [17] with Fourier features [26]	Latent code supplied by the contour encoder instead of per-shape optimisation
Output coupling	Surfaces as zero level sets of signed-distance fields [17]	Epicardial field defined as the endocardial field minus a strictly positive offset δ, which a wall-distance loss supervises directly
Training	Surface and distance fitting [17]; Eikonal regularisation [25]	Two stages, synthetic SSM pre-training then clinical ED/ES fine-tuning, through one shared sample format

synthetic meshes are used for pre-training only and never appear in the validation or test sets, so every number reported in Chapter 4 is computed on clinical cases. The role of each stream in the two training stages is summarised in Chapter A. With every case in one format, the model can now be described without reference to the source of a sample.

3.3. Model Architecture

The model combines a global contour encoder, a local feature grid, and an implicit decoder (Figure 3.7). The encoder summarises the overall ventricular shape, while the grid retains local contour information. The decoder uses both descriptions to predict the endocardial field and a positive offset defining the epicardial field. Tissue and cardiac-phase labels allow the same model to process both boundaries at ED and ES.

Its design builds on established methods for learning from point sets and representing shapes implicitly. Contours are treated as an unordered point set and summarised by a shared per-point network with max-pooling, as in PointNet [38]. Spatial detail lost in that summary is recovered from a convolutional feature grid sampled at each query point, following Convolutional Occupancy Networks [48]. The surfaces are zero level sets of signed-distance fields predicted by a latent-conditioned network in the manner of DeepSDF [17], with Fourier features of the query coordinates [26] and Eikonal regularisation during training [25].

Table 3.3 summarises the adaptations made for sparse LV contours and paired-surface reconstruction. The following subsections define the inputs, encoder, and decoder.

Compared with the recent cardiac implicit models reviewed in Section 2.3.2, which reconstruct from image data [19] or from sparse segmentations of both ventricles with long-axis views and a learned biventricular prior [20], the proposed model takes only LV contour points with tissue and phase labels, and it enforces the nesting of the two surfaces by construction.

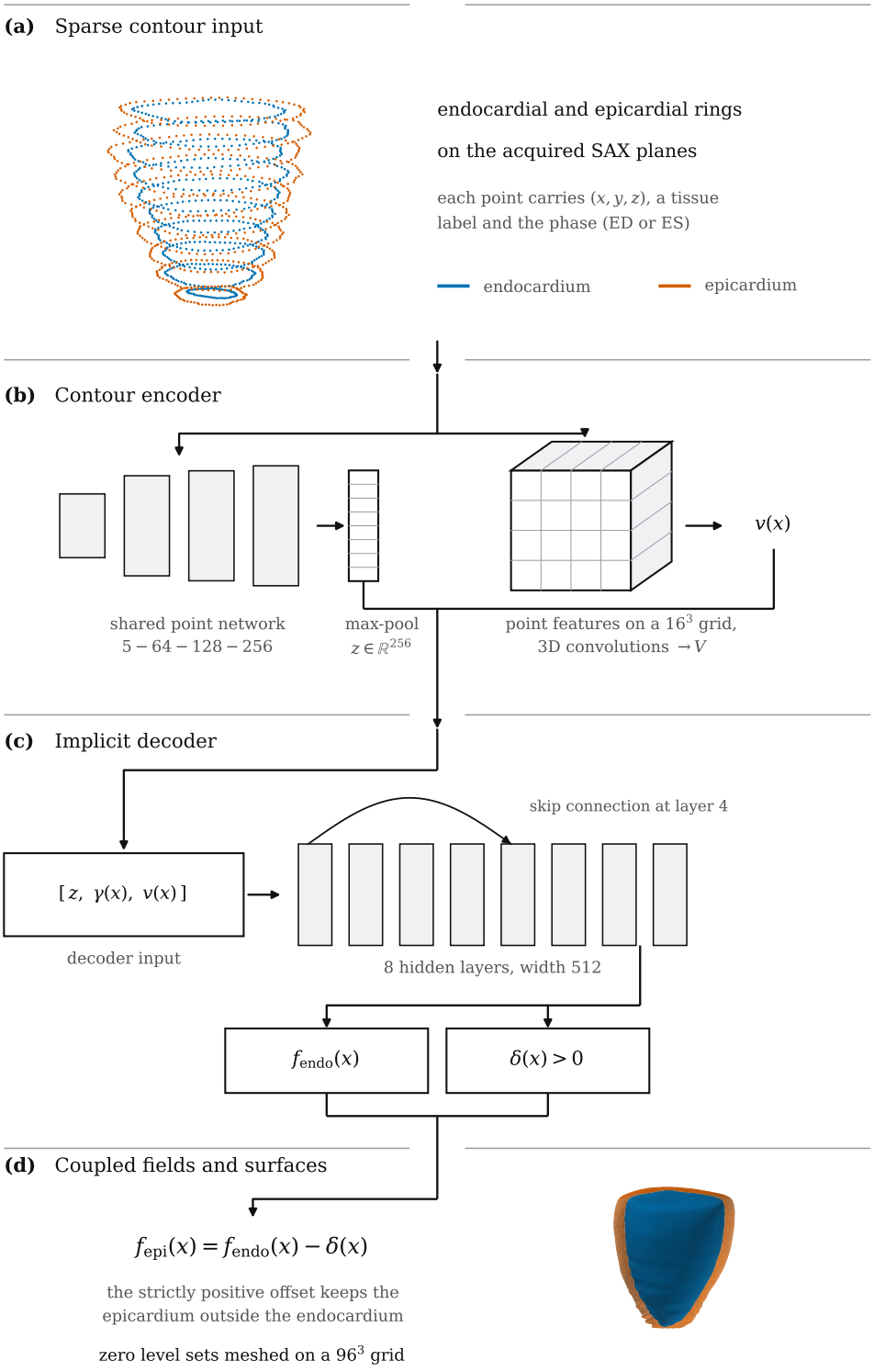


Figure 3.7. Phase-conditioned model architecture, from the sparse contour observation (a) through the global and local encoder paths (b) and the implicit decoder (c) to the coupled fields and extracted surfaces (d). The point cloud in (a) and the cut surfaces in (d) are the observed contours and the corresponding reconstruction of one ACDC case, with the epicardium opened to expose the endocardial cavity. The offset δ shown in the figure is a separation between field values, not a measured thickness.

3.3.1. What the model observes

Reconstruction begins with the contour points prepared in the previous section, not with the MRI images themselves. Each point tells the model where a boundary was observed, which boundary it belongs to, and at which cardiac phase. The notation below describes how this information is supplied to the network.

For a batch of B cases processed together, the encoder receives the contour points as the padded tensor defined in Equation (3.6):

$$P \in \mathbb{R}^{B\times N\times 5} \tag{3.6}$$

and a validity mask $M \in \{0,1\}^{B\times N}$ that marks which of the N rows are real points; padding only allows cases with different numbers of points to share a batch. Each valid row is $p_i = (x_i, y_i, z_i, t_i, q_i)$: normalised spatial coordinates, a tissue label t_i that distinguishes endocardial from epicardial points, and a phase label q_i that identifies ED or ES. The network receives neither image intensities nor a template mesh; everything it knows about a case comes from these five values per point. Attaching the phase to every point, rather than introducing it only at the decoder, lets both encoders interpret the same arrangement of contours differently for a diastolic and a systolic heart. The tissue label serves a different purpose: it allows evidence about the cavity boundary and evidence about the outer boundary to be summarised separately before they are combined.

3.3.2. Contour encoder

Before predicting a surface, the model needs a useful description of the observed contours. The encoder builds this description at two scales: a summary of the whole ventricle and a spatial grid of local contour information. The decoder can then use both the overall shape and the nearby observations when predicting a boundary between slices.

Following PointNet [38], the global pathway applies a shared network with layer widths $5 \to 64 \to 128 \to 256 \to 256$ and rectified linear unit (ReLU) activations to each point independently. The encoder therefore does not depend on the order of points within a ring or between slices. For the points $P = \{p_i\}_{i=1}^{N}$ of one case, the network assigns each point a 256-dimensional feature, as expressed in Equation (3.7):

$$h_i = \phi_{\text{enc}}(p_i), \tag{3.7}$$

where ϕ_{enc} denotes the shared network.

Unlike standard PointNet, this pooling operation takes separate maxima for the endocardial set $\mathcal{E}$ and the epicardial set $\mathcal{P}$, as shown in Equation (3.8):

$$z_{\text{endo}} = \max_{i\in\mathcal{E}} h_i, \qquad z_{\text{epi}} = \max_{i\in\mathcal{P}} h_i, \tag{3.8}$$

where the maximum is computed independently in each feature channel. This preserves the strongest response to each learned pattern for each boundary before their information is combined. The resulting descriptors form a 512-dimensional vector, which is projected to the global latent code according to Equation (3.9):

$$z = W_z[z_{\text{endo}}, z_{\text{epi}}] + b_z, \tag{3.9}$$

where $z \in \mathbb{R}^{256}$ provides the decoder with a compact description of the complete case.

Compressing all points into z discards their locations. A local pathway, following Convolutional Occupancy Networks [48], retains this spatial information. A second shared network with widths $5 \to 64 \to 64 \to 32$ assigns a 32-dimensional feature to every valid point. These features are placed in a regular 16^3 grid over the normalised extent $[-1.8, 1.8]^3$. Features that fall in the same cell are averaged, preventing densely sampled regions from having greater influence solely because they contain more points.

Three $3 \times 3 \times 3$ convolutions then propagate information between neighbouring cells. Group Normalisation [49] and ReLU follow the first two convolutions, producing the feature volume in Equation (3.10):

$$V \in \mathbb{R}^{32 \times 16 \times 16 \times 16}. \tag{3.10}$$

For a query point x, trilinear interpolation of the eight surrounding cells gives the local feature defined in Equation (3.11):

$$v(x) = \mathsf{Interp}(V, x). \tag{3.11}$$

The global code z is shared by every query in a case, whereas $v(x) \in \mathbb{R}^{32}$ describes the contour evidence near x. The grid is deliberately coarse because a few hundred contour points cannot support a fine volume. Its purpose is to provide local context to the decoder, not to define the reconstruction resolution.

3.3.3. Implicit decoder with positive-offset coupling

Although the encoder describes the case, it does not yet produce a surface. The decoder uses that description to predict field values at locations throughout the volume, including between the observed slices. The locations where each field is zero form its surface; coupling the two fields keeps the inner and outer boundaries separated.

Each surface is represented by the decoder as the zero level set of a function of continuous coordinates, in the manner of DeepSDF [17]. A network fed with plain coordinates tends towards overly smooth fields, so each query point is first expanded with Fourier features [26], which add sine and cosine components at several spatial frequencies. For $x \in \mathbb{R}^3$, the encoding is defined in Equation (3.12):

$$\gamma(x) = [x, \mathsf{sin}(2^0 \pi x), \mathsf{cos}(2^0 \pi x), \ldots, \mathsf{sin}(2^{L-1} \pi x), \mathsf{cos}(2^{L-1} \pi x)], \tag{3.12}$$

with $L = 3$, giving $3 + 6L = 21$ components. The encoded coordinate is concatenated with the global code to form the decoder input in Equation (3.13):

$$h_{\mathsf{in}}(x) = [z, \gamma(x)] \in \mathbb{R}^{277}. \tag{3.13}$$

An input layer maps this vector to width 512, and the main body of the decoder, its trunk, applies eight fully connected layers of the same width with Softplus activations, smooth non-linear functions that keep the field smooth and differentiable. At the fourth hidden layer, the 277-dimensional input is appended to the current state and projected back to width 512; this skip connection lets the deeper layers see the case code and the coordinate directly rather than through the whole trunk. The local feature $v(x)$ enters

the decoder at the input layer, at this skip connection, and in the offset head described next, so that the prediction at a point depends on the nearby contours as well as on the global shape.

At the output stage, the model departs from an ordinary implicit decoder. Predicting the endocardium and the epicardium as two unrelated fields would let them cross or touch wherever the data are sparse, producing a wall of zero or negative thickness between slices. The final hidden representation therefore feeds two coupled output layers, or heads: one predicts the endocardial signed-distance value $f_{\text{endo}}(x)$, the other a strictly positive offset $\delta(x)$. The epicardial field is coupled to these outputs through Equation (3.14):

$$f_{\text{epi}}(x) = f_{\text{endo}}(x) - \delta(x), \qquad \delta(x) > 0. \tag{3.14}$$

Positivity is enforced in the implementation by a sigmoid-based mapping, which squashes any real number into a bounded positive range, with additional headroom for large offsets. Because the epicardial field is always lower than the endocardial field by δ, its zero level set always lies outside the endocardial one, which rules out wall collapse by construction. Two qualifications apply. The constraint does not by itself guarantee an anatomically valid surface or a valid mesh once the field has been sampled on a grid, which is why the extracted meshes are checked and repaired in Section 3.4.2. And δ is a difference between two field values at one point, not the distance between the two extracted surfaces, so it is not a measurement of wall thickness; thickness is measured on the extracted surfaces with the methods of Section 3.5. How the two fields are turned into meshes is described in Section 3.4.2; the complete configuration of the model is listed in Table A.4 of Chapter A.

3.4. Training and Inference

The same architecture and loss are used for synthetic pre-training and clinical fine-tuning. The trained fields are then converted into meshes for reconstruction and thickness evaluation.

3.4.1. Two-stage training and loss

In the first stage the model is trained only on the 1300 synthetic ED cases, which give it a broad prior for LV shape under controlled conditions and tolerate stronger augmentation of the sparse observations. In the second stage it continues from the pre-trained weights on the ACDC and M&Ms-2 cases of both phases, adapting to clinical contour quality and to systolic anatomy. The patient-level split and the augmentation rule of Section 3.2.3 apply unchanged.

One phase-conditioned model is trained rather than one model per phase. The reason follows from the data. Because the SSM provides ED geometry only, a separate ES model could not have received equivalent pre-training, whereas a shared model lets the ES cases benefit from the prior learned on the synthetic ED corpus, with the phase label telling the encoder which configuration it is seeing. The RBF-derived targets are constructed identically in both phases, so no second asymmetry is introduced at fine-tuning. The cost is that phase information is coarse, a binary label rather than a position in the cardiac cycle, and that the prior remains biased towards diastolic shape.

Five complementary constraints shape the objective. The surface term places the predicted zero level sets on the target surfaces, while the dense SDF term fits the stored signed distances throughout

the sampled volume. The Eikonal term encourages unit gradient magnitude, as required of a signed-distance field, and the off-surface term prevents additional zero level sets in free space. Finally, the wall term fits δ to the target separation between the endocardium and epicardium. This final constraint teaches the coupling to represent wall separation rather than merely preserve the ordering of the two fields. The first four terms follow established implicit-surface objectives [17, 25]; the wall term is specific to the coupled output.

Formally, let $\mathcal{E}$ and $\mathcal{P}$ denote sampled endocardial and epicardial surface points, $\mathcal{Q}$ the stored query points, $\mathcal{R}$ the union of near-surface and free-space points used for the gradient term, and $\mathcal{F}$ the free-space subset. At each $q \in \mathcal{Q}$, $y_{\text{endo}}(q)$ and $y_{\text{epi}}(q)$ are the stored signed-distance targets, and at each $x \in \mathcal{E}$ the wall target $t(x) = -y_{\text{epi}}(x)$ is the distance from that endocardial point to the epicardial surface. Writing $\mathbb{E}$ for the mean over the indicated set of points, the five terms are defined in Equation (3.15):

$$\begin{aligned}
\mathcal{L}_{\text{surf}} &= \mathbb{E}_{x\in\mathcal{E}}[|f_{\text{endo}}(x)|] + \mathbb{E}_{x\in\mathcal{P}}[|f_{\text{epi}}(x)|],\\
\mathcal{L}_{\text{SDF}} &= \mathbb{E}_{q\in\mathcal{Q}}[|f_{\text{endo}}(q) - y_{\text{endo}}(q)|] + \mathbb{E}_{q\in\mathcal{Q}}[|f_{\text{epi}}(q) - y_{\text{epi}}(q)|],\\
\mathcal{L}_{\text{eik}} &= \mathbb{E}_{x\in\mathcal{R}}[(\|\nabla_x f_{\text{endo}}(x)\|_2 - 1)^2] + \mathbb{E}_{x\in\mathcal{R}}[(\|\nabla_x f_{\text{epi}}(x)\|_2 - 1)^2],\\
\mathcal{L}_{\text{off}} &= \mathbb{E}_{x\in\mathcal{F}}[e^{-\alpha|f_{\text{endo}}(x)|}] + \mathbb{E}_{x\in\mathcal{F}}[e^{-\alpha|f_{\text{epi}}(x)|}],\\
\mathcal{L}_{\text{WT}} &= \mathbb{E}_{x\in\mathcal{E}}[\ell_{\beta_{\text{WT}}}(\delta(x) - t(x))],
\end{aligned} \tag{3.15}$$

where $\alpha = 5$ and ℓ_β is the Smooth-L1 penalty [50], which grows like a squared error for small differences and like an absolute error for large ones, so that a few badly wrong points do not dominate the term. The penalty is defined in Equation (3.16):

$$\ell_\beta(a) = \begin{cases} a^2/(2\beta), & |a| < \beta, \\ |a| - \beta/2, & |a| \geq \beta, \end{cases} \qquad \beta_{\text{WT}} = 0.02. \tag{3.16}$$

All distances are expressed in the normalised coordinate system of each case, and the terms are combined into the complete objective in Equation (3.17):

$$\mathcal{L}_{\text{total}} = \lambda_{\text{surf}}\mathcal{L}_{\text{surf}} + \lambda_{\text{SDF}}\mathcal{L}_{\text{SDF}} + \lambda_{\text{eik}}\mathcal{L}_{\text{eik}} + \lambda_{\text{off}}\mathcal{L}_{\text{off}} + \lambda_{\text{WT}}\mathcal{L}_{\text{WT}}, \tag{3.17}$$

where the weights λ balance the five terms. The Eikonal term is computed in full precision to keep its gradient calculation stable; the weights and implementation settings are reported in Table A.3 of Chapter A.

The reported run uses the AdamW optimiser [51], a standard gradient-based method, with a learning rate (step size) of 2×10^{-5} and weight decay 5×10^{-4}; the learning rate follows cosine annealing with warm restarts [52], decreasing along a cosine curve and being reset at intervals, with an initial restart period of 100 epochs (passes over the training set). Pre-training lasted approximately 200 epochs; fine-tuning then ran for 776 epochs, and the checkpoint at epoch 695 was selected by validation loss. Training uses mixed precision (lower-precision arithmetic where it is safe) and gradient clipping at 1.0 (a cap on the size of each update); each of two GPUs processes a batch of 8 cases and gradients are accumulated over two steps, for an effective batch size of 32. Weights are lightly clipped instead of spectrally normalised,

because the decoder is evaluated at many points per step and GPU memory is the limiting resource. Table A.5 in Chapter A collects these settings.

3.4.2. Inference, offset extraction, and comparators

At inference time, the encoder turns the contours into z and V once, and the decoder is evaluated on a regular 96^3 grid of points. Marching cubes extracts the zero level sets of f_{endo} and f_{epi} from this grid as the raw endocardial and epicardial meshes [24]; the mesh is thus a discretisation chosen at inference time, not a property of the architecture, and could be extracted at any resolution. For evaluation, faces of zero area are removed, only the largest connected piece of each mesh is kept, holes are closed, and, where needed, a mesh is rebuilt from a cleaned inside/outside voxel mask. This repair produces usable geometry for surface and thickness measurement, but it also means that the reported metrics describe a repaired mesh rather than the unmodified decoder output.

Because the decoder predicts δ at every point, a raw offset is also available at every endocardial vertex. Its value in millimetres is given by Equation (3.18):

$$o_{\text{raw}}(x) = s\,\delta(x, z, v(x)), \tag{3.18}$$

where s is the scale that maps normalised coordinates back to millimetres. For visualisation only, this offset is calibrated separately for each phase as $o_{\text{cal}} = \alpha\, o_{\text{raw}}$, where α is the ratio of the mean of a reference thickness, computed directly from the segmentation voxels with the distance-transform method of Section 3.5.1, to the mean raw offset. The calibration fixes the mean by construction and is not a test of accuracy. As established in Section 3.3.3, all reported thickness values are instead computed on the extracted surfaces using the methods described in Section 3.5.

The model is compared with two reconstructions built from exactly the same contour rings. These comparators embody different and well-understood priors. The first is an RBF implicit fit obtained with the procedure of Section 3.2.2: it interpolates the rings through a smooth field and belongs to the same family as the fine-tuning targets. The second is a fit of the UK Digital Heart Project shape model to the rings. The mean shape is first aligned to the rings by long axis, scale, and rotation about the long axis, where the long axis is the line through the centres of the rings; the fit then alternates three steps, matching each ring point to its nearest model vertex, re-aligning the model to the rings by translation, rotation, and scale, and re-estimating the first 25 mode coefficients with the coefficients constrained to three SDs. As the published model provides ED modes only, this comparator exists at ED alone. Both pass through the same repair and smoothing as the model output.

None of the three reconstructions is treated as ground truth. Evaluation compares their distances from the input contours and their pairwise surface and volume agreement. These comparisons assess reconstruction consistency, not the contribution of individual model components.

3.5. Wall-Thickness Evaluation

The preceding section produces the endocardial and epicardial surfaces used for thickness measurement. Measuring the distance between them requires pairing each point on one surface with a point on

the other. On a curved, tapering wall, the result depends on the pairing rule: for example, it may use the nearest point, travel perpendicular to the surface, or follow a smooth path through the wall. These rules can give different thickness values on the same geometry. Four established methods are therefore compared on the same reconstructed surfaces. The segmentation volume supplies the contour rings and the orientation of the AHA coordinate system; thickness is measured only on the reconstructed surfaces. The method whose measurements are most consistent in Chapter 4 is used for the regional analysis. This evaluation compares measurement methods and produces three-dimensional thickness maps; it does not assess diagnostic performance for infarction, scarring, or hypertrophy.

3.5.1. Selected thickness methods

These four methods span the main families identified in Section 2.3.3: distance fields computed on a voxel grid, rays cast from one surface to the other, and methods based on PDEs, which define a smooth field across the wall and measure along it (Table 3.4). None is assumed in advance to be the best. Each returns a local thickness in millimetres at the endocardial vertices or at points projected onto the endocardium, and the resulting fields are summarised by their mean, median, SD, 5th and 95th percentiles, fraction of valid values, and runtime, and shown as three-dimensional colour maps and as AHA-17 bullseye plots, circular diagrams with one cell per segment. Figure 3.8 contrasts the four pairing rules on one wall cross-section.

Table 3.4. Wall-thickness methods selected for evaluation on the model-generated LV geometry, with the EDT boundary sum kept as the simplest baseline.

Method	Category	Main measurement principle
Laplace field	PDE	Transmural field lines from a harmonic potential
Yezzi–Prince	PDE	Eulerian transport equations along the transmural field
SDF cone rays	Ray / SDF	Median of multiple cone rays around the surface normal
EDT boundary sum	Volumetric	Sum of endocardial and epicardial distance transforms

EDT boundary sum is the simplest method. A distance transform assigns to every voxel of the myocardium its straight-line distance to the nearest point of a given boundary; the method computes one such transform for each surface and adds them, as defined in Equation (3.19):

$$t^{\mathrm{EDT}}(x) = D_{\mathrm{endo}}(x) + D_{\mathrm{epi}}(x), \tag{3.19}$$

where D_{endo} and D_{epi} are the distance transforms of the endocardium and the epicardium. The sum is exact for two parallel flat walls, but it is limited by voxel size and slightly overestimates thickness where the wall is strongly curved, because the two nearest points do not lie on a common path across the wall.

By contrast, the Laplace field method replaces nearest points with a smooth path across the wall. A scalar field ψ is defined over the myocardium by solving Laplace's equation with the boundary conditions in Equation (3.20):

$$\nabla^2 \psi = 0, \qquad \psi|_{\mathrm{endo}} = 0, \qquad \psi|_{\mathrm{epi}} = 1, \tag{3.20}$$

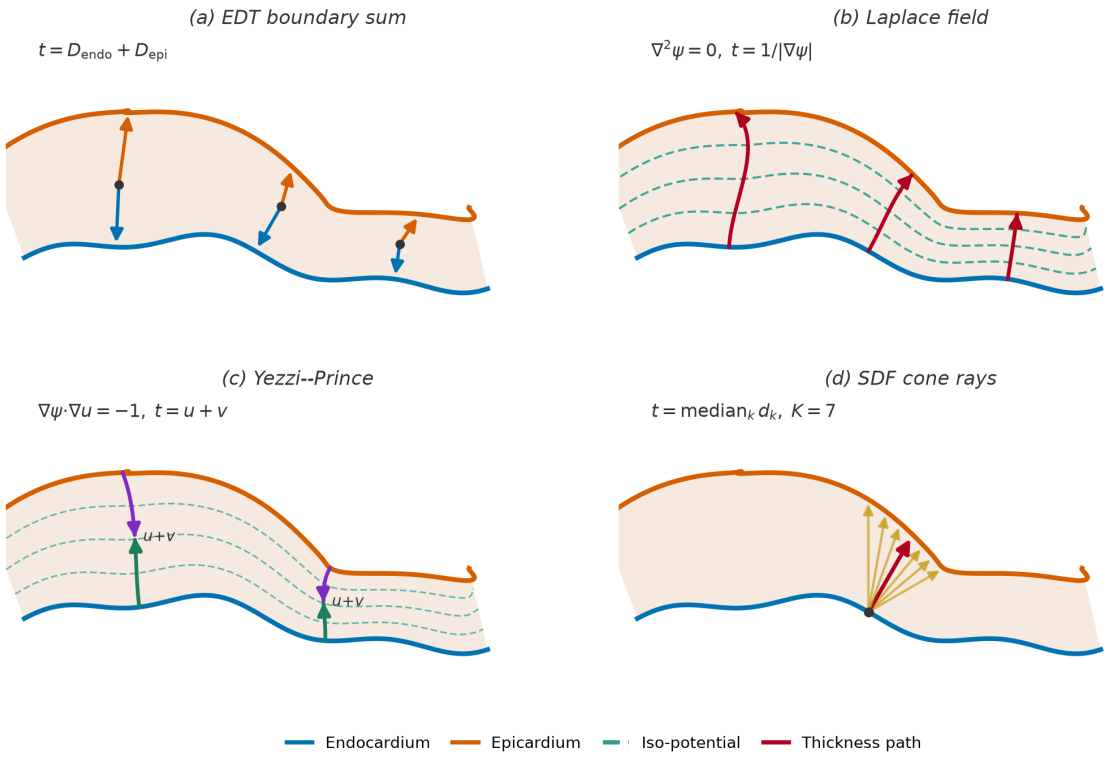


Figure 3.8. Conceptual comparison of the four wall-thickness methods on a single myocardial cross-section (endocardium in blue, epicardium in orange). (a) EDT boundary sum: distances from an interior point to each boundary are added. (b) Laplace field: curved paths of a potential rising from 0 to 1 cross its level lines (dashed) at right angles, and thickness follows from the gradient magnitude. (c) Yezzi–Prince: travel distances u and v from opposite boundaries along the same field are summed. (d) SDF cone rays: the median distance of a fan of rays around the endocardial normal is kept (highlighted), limiting the effect of any single ill-oriented ray.

so that ψ rises smoothly from 0 on the endocardium to 1 on the epicardium, each value being the average of its neighbours (a harmonic field), and acts as a potential whose field lines run across the wall. Thickness is then estimated from the magnitude of the gradient of ψ, the rate at which the field changes across the wall, which is small where the wall is thick and large where it is thin. The paths that follow the gradient never cross one another, which is why the method, introduced for the thickness of the brain cortex by Jones et al. [44], is widely used for layered tissue; its weakness is that the numerical estimate can degrade where the wall is thin or the gradient is small. The Yezzi–Prince method [45] removes this weakness by computing, along the same field, the distance travelled from each boundary with a pair of transport equations solved on the voxel grid (an Eulerian formulation, computed at fixed grid points rather than by tracing individual paths). Their simplified form is given in Equation (3.21):

$$\nabla\psi \cdot \nabla u = -1, \qquad t = u + v, \tag{3.21}$$

where u and v are the distances travelled from the two boundaries and their sum is the thickness.

For cone-ray estimation, each endocardial point is paired with the epicardial point reached along its surface normal, while multiple rays guard against a single badly oriented direction. Following the shape diameter function of Shapira et al. [42], K rays are cast within a cone of half-angle 30° around the normal,

and the median of the valid intersection distances is kept, as defined in Equation (3.22):

$$t_i^{\text{cone}} = \text{median}_{k=1}^{K}\, d_{ik}, \qquad K = 7. \tag{3.22}$$

As a result, the estimate remains approximately transmural, measuring across rather than along the wall, while being more robust than a single normal ray.

What is specific to this work is not the methods themselves but how they are used: all four are run on the same reconstructed surfaces under the same conditions, and their agreement with one another, rather than with a reference that does not exist, decides which one carries the regional analysis.

3.5.2. Local three-dimensional wall thickness

Before any regional summary, each method produces a thickness value at every endocardial vertex of the repaired meshes of Section 3.4.2, that is, a continuous map over the whole reconstructed wall rather than a handful of measurements on selected slices. This map is the primary output of the measurement stage. It shows the gradual change from the thin apex to the thicker base and preserves local thinning or thickening that averaging within an AHA segment would hide, and a fixed colour scale makes the maps comparable across patients and phases (Figure 3.9).

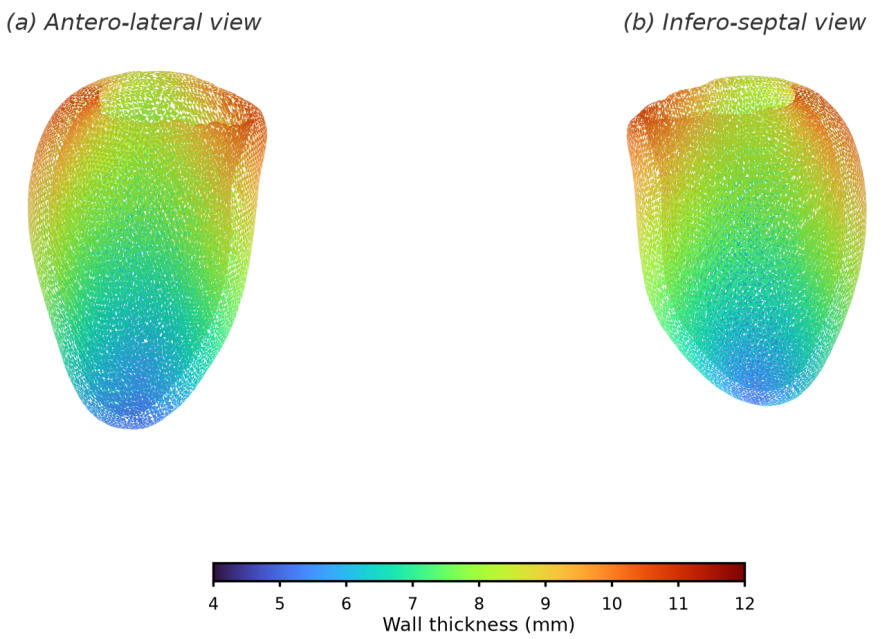


Figure 3.9. Local three-dimensional wall-thickness map on the reconstructed LV endocardial surface, shown from two opposing viewpoints. Each vertex is coloured by its local wall thickness on a fixed millimetre scale, so that the continuous spatial variation from the thin apex to the thicker base is visible directly on the geometry rather than only as regional averages.

3.5.3. Regional AHA-17 mapping

For clinical reporting, the local map from the selected method is summarised in the AHA 17-segment model [11]. The model was defined for two-dimensional imaging and divides the ventricle along its long

axis into basal, mid-cavity, and apical thirds. A representative SAX slice from each third is partitioned into angular sectors around the cavity: six sectors in the basal and mid-cavity regions and four in the apical region. The sectors are numbered from the anterior wall, with the septal segments facing the RV, while the apical tip forms the seventeenth segment. In routine use, an observer measures the thickness of each segment on the corresponding slice.

The local thickness map contains many more values than a regional report needs. Assigning those values to anatomically oriented regions allows the same parts of the ventricle to be compared across cases and cardiac phases, while retaining the full surface map for more detailed inspection.

Transferring this definition to a reconstructed surface requires only that every vertex receive the two coordinates the definition relies on, a position along the long axis and an angle around it; the segments then become patches of the three-dimensional surface rather than readings on single slices. The long axis is taken as the direction of the SAX stack. The base and apex are the first and last slices that contain myocardium, the base being the end whose cavity cross-section is larger, and every endocardial vertex is given a normalised position between 0 at the base and 1 at the apex. Vertices in the first third of this range form the basal ring, those in the second third the mid-cavity ring, those between two thirds and 0.85 the apical ring, and those beyond 0.85 the apical cap, which is segment 17.

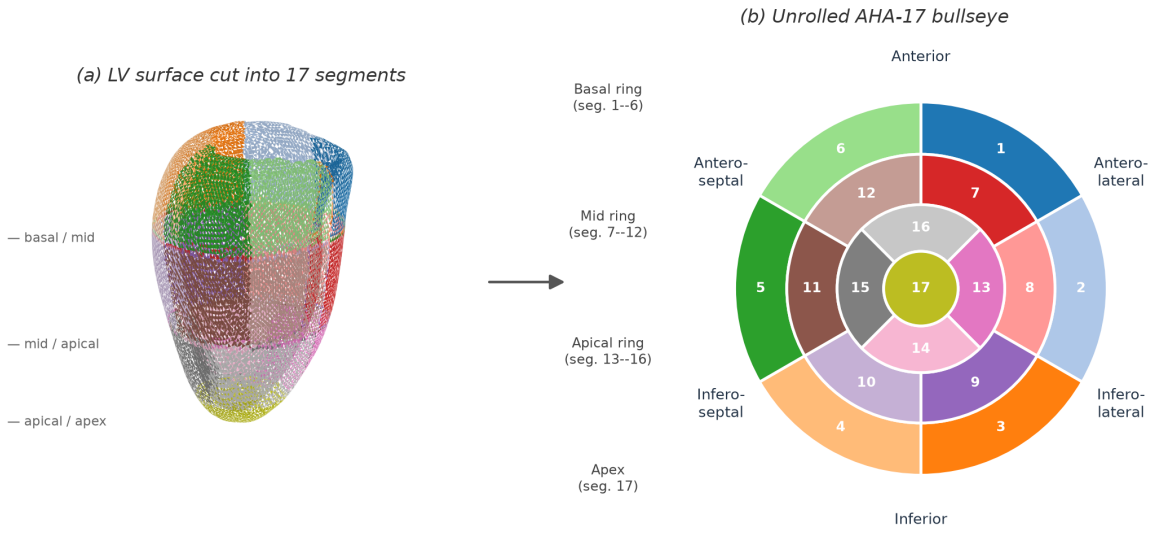


Figure 3.10. AHA-17 partitioning of the reconstructed LV surface and corresponding bullseye representation.

The angle of each vertex is measured in the SAX plane around the centre of the endocardial vertices. Its orientation is fixed by the right-ventricular label: the direction from the LV centre to the right-ventricular centre identifies the septum, and the anterior reference lies a quarter turn from it. The basal and mid-cavity rings are divided into six sectors of $60°$, while the apical ring is divided into four sectors of $90°$. These divisions form segments 1–6, 7–12, and 13–16, respectively (Figure 3.10).

Before assigning the segments, a quality-assurance view displays the detected base, apex, and septal direction. The rings are then unrolled onto concentric circles to form the bullseye, with the basal

ring outermost and the apical cap at the centre. Each cell therefore summarises the vertices of one surface patch. Within each segment, the mean and percentiles provide complementary regional summaries rather than relying on a single statistic. These summaries complement rather than replace the per-vertex maps.

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CHAPTER 4

Results and Evaluation

This chapter presents the experimental evaluation of the approach described in Chapter 3.

4.1. Experimental Setup

The evaluation uses the final model, which was pre-trained on the synthetic SSM meshes and then fine-tuned on the real ED and ES data, as described in Chapter 3. Datasets and patient-level splits are reported in Section 3.2.3. General reconstruction and wall-thickness evaluations use the held-out test patients from both ACDC and M&Ms-2. The analysis by disease group uses a separate ACDC subset of 30 test patients, comprising 20 normal (NOR) and 10 HCM cases. The HCM group is included so that the model can be examined on abnormal as well as normal LV geometry. No independent 3D ground truth is available, so the following results describe contour fidelity and agreement between methods rather than reconstruction accuracy. Three questions guide the evaluation: how faithfully the model follows the observed LV contours and how far it agrees with alternative reconstructions of them, whether it represents the ED–ES change while maintaining a valid wall, and how closely the resulting thickness measurements agree with those obtained from an independent method on the same input.

4.1.1. Metrics

Reconstruction geometry is characterised with complementary surface and volume quantities, following the metric framework of Taha and Hanbury [21] and the conventions used for cardiac segmentation [1] and for implicit-surface learning [18]. These quantities are used in two roles: between a reconstruction and the contour rings that produced it, and between two reconstructions of the same rings. In the first role they measure fidelity to the observations; in the second they measure disagreement between methods, and are symmetric, so they belong to neither method as an error.

Chamfer distance is the average distance between two surfaces, taken in both directions. It is reported for the endocardium and the epicardium in millimetres. The *contour distance* is the corresponding one-sided quantity from each observed ring point to the reconstructed surface, summarised by its mean and its 95th percentile.

Dice similarity coefficient (DSC) measures the solid rather than the surface. After converting both surfaces to voxel grids, it is twice the shared volume divided by the sum of the two volumes. It is computed for the endocardial cavity and for the myocardium and lies between zero and one. Enclosed *volumes* are reported in millilitres for each method separately. A volume ratio is deliberately not reported: it would require designating one reconstruction as correct, which the available data do not support.

The watertight rate is the fraction of reconstructions that form a closed surface without holes. It is stated in the text as a precondition rather than as a score, because a mesh that is not closed has no volume for the DSC to use.

HD95, the 95th-percentile Hausdorff distance, describes the tail of the distribution and is reported with the Chamfer distance. The average symmetric surface distance is omitted because it tracked the Chamfer distance to within 0.06 mm throughout. The intersection over union is omitted because it is a direct transformation of the DSC; the F-score depends on one selected tolerance, and normal consistency measures orientation rather than positional agreement.

Wall thickness is summarised by the mean, SD, and 5th and 95th percentiles across the myocardial surface, all in millimetres. The percentiles describe the thin apical and thick basal extremes without being distorted by isolated outliers. When the two cardiac phases are compared, the absolute thickening $\Delta_{\mathrm{ES}-\mathrm{ED}}$ and the relative thickening as a percentage of the ED value are also reported. These are the standard descriptors of regional contractile function.

The four wall-thickness methods are compared on the reconstructed geometry, using the agreement framework of Bland and Altman [53] instead of a comparison of mean values. The reported quantities are the *Pearson correlation coefficient* r, the *mean absolute error* (MAE) and *root-mean-square error* (RMSE) in millimetres, and the Bland–Altman *bias* with its *95% limits of agreement* (LoA) (bias $\pm 1.96\,\sigma$). Absolute consistency is summarised by the two-way *intraclass correlation coefficient* (ICC), read against the thresholds of Koo and Li [54]. These are reported together because correlation alone does not show agreement: two methods can rise and fall together and still differ by a constant amount, which only the bias reveals.

4.1.2. Methods Used for Comparison

The reconstruction is compared with two alternative reconstructions of the same input contour rings: a radial-basis-function implicit fit, which interpolates the rings through a smooth field, and a fit of the public SSM, which keeps the reconstructed shape within the variation represented by the population model. These methods make different assumptions about the shape between slices, so differences between their results show how strongly those assumptions affect the reconstructed geometry. Both use the same rings as the model and undergo the same mesh repair and measurement steps, as described in Section 3.4.2, so their numbers are directly comparable. They are evaluated only at ED: the public SSM is distributed with ED modes only.

The RBF fit is used for comparison, not as ground truth. It belongs to the same family as the surfaces fitted to the contours for training, so close agreement with the model is partly expected and is not independent confirmation. By contrast, the SSM fit is independent of the training targets, but some patients have shapes outside the variation represented by its training population. Wall thickness is compared between the model, the RBF fit, and the SSM fit with the four algorithms selected in Section 3.5.1: the Laplace field method, the Yezzi–Prince method, the SDF cone-ray method, and the EDT boundary sum. Because the SAX stacks are sparse relative to their in-plane resolution, all three reconstruction methods must infer the through-plane geometry, and their absolute millimetre values should be read with care.

4.2. Training

Training proceeds in two stages. The first pre-trains the signed-distance decoder for roughly 200 epochs on the synthetic SSM meshes. This stage provides a geometric prior, and its checkpoint initialises the second stage. The reported model is then fine-tuned on the real ED and ES data described in Chapter 3. The available training history records the fine-tuning stage; no numerical claim is made about the pre-training loss because its per-epoch log was not retained.

Fine-tuning runs for 776 epochs with the AdamW optimiser (learning rate 2×10^{-5}, weight decay 5×10^{-4}) under a cosine-annealing schedule with warm restarts, mixed-precision arithmetic, and an effective batch size of 32. The composite objective is the five-term loss of Equation (3.17), combining surface, dense SDF, Eikonal, off-surface, and wall-thickness supervision. Figure 4.1 plots the total loss and its principal components together with two optimisation diagnostics. The total training loss decreases from 10.07 at the first fine-tuning epoch to 0.91, and the validation loss settles at 0.70, with the best checkpoint selected at epoch 695 before mild over-fitting appears. The surface and Eikonal terms fall by more than an order of magnitude, indicating successful optimisation of the surface-fitting and Eikonal objectives. Consistently, the mean gradient norm $\|\nabla f\|$ stabilises at 0.996, close to the unit value encouraged by the Eikonal term.

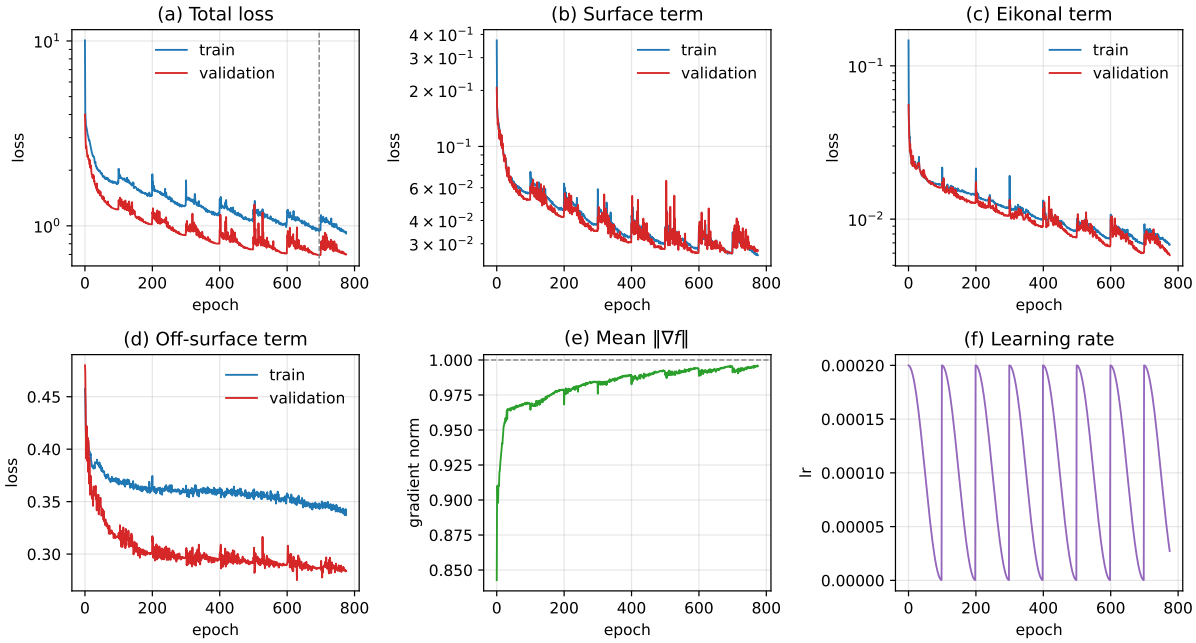


Figure 4.1. Training history. (a)–(d) Total loss and its surface, Eikonal, and off-surface components for the training and validation splits; the dashed line marks the best epoch. (e)–(f) Mean gradient norm (target one) and cosine-annealing learning rate. Synthetic pre-training is not shown.

4.3. Results

Three analyses cover reconstruction consistency, wall-thickness measurement, and regional analysis with the AHA 17-segment model.

4.3.1. Reconstruction Consistency

Contour fidelity measures how closely each surface follows its input rings, and pairwise agreement measures how much the three reconstructions differ from one another. All values below are at ED and cover the held-out ACDC and M&Ms-2 test cohort. Every surface from every method is watertight after the shared repair stage.

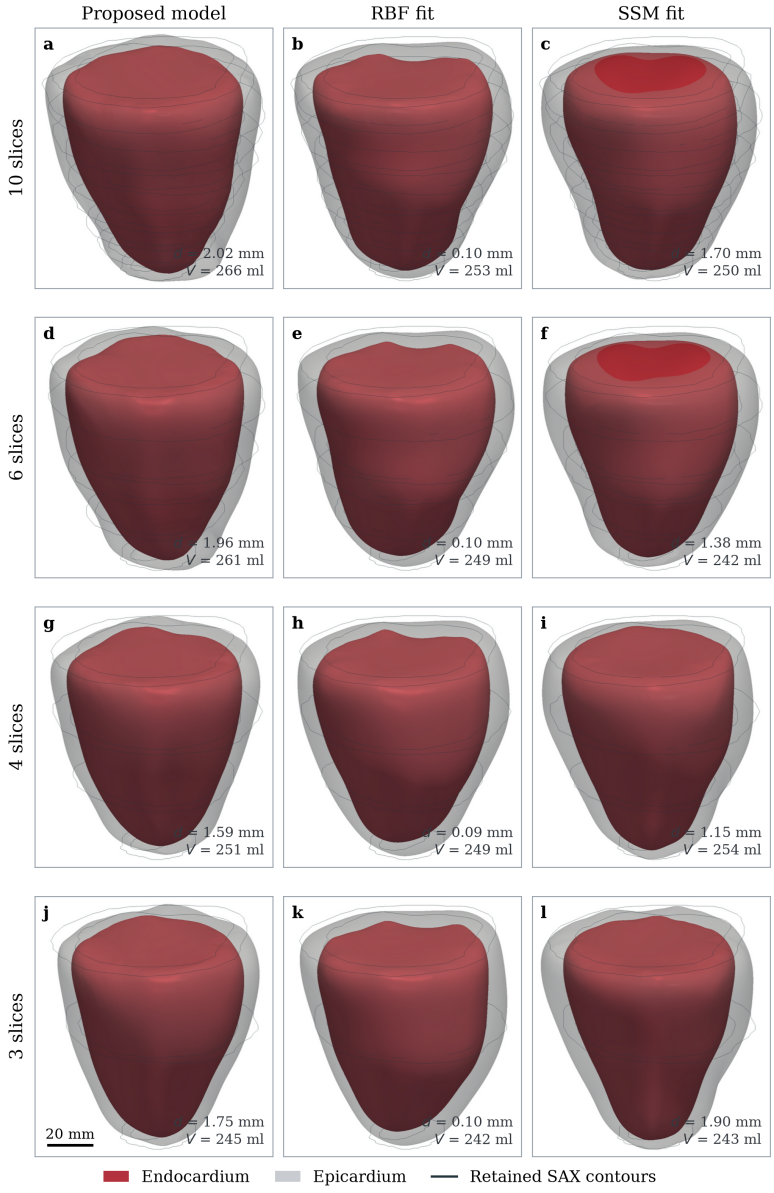


Figure 4.2. Left-ventricular surfaces of one ACDC case reconstructed from an ED SAX stack with ten, six, four, and three retained slices (rows) by the proposed model, RBF fit, and SSM fit (columns). Red: endocardium; grey: epicardium; dark rings: input contours.

Figure 4.2 shows one case reconstructed by the three methods. In the complete ten-slice row, the three surfaces share the same overall ventricular shape. The reduced-slice rows are analysed in Section 4.3.2.

Table 4.1 reports the distance from each input contour ring to the reconstructed surface and the enclosed volumes.

Table 4.1. Distance from the input SAX contour rings to each reconstructed surface, and enclosed volumes, at ED over the held-out test cohort (mean $\pm$ SD of per-patient values; p_{95}: mean per-patient 95th percentile). Bold marks the lowest distance.

Method	Surface	Contour distance (mm)	p_{95} (mm)	Volume (ml)
Proposed model	Endocardium	1.50 ± 0.23	3.56	119.3 ± 31.3
	Epicardium	1.96 ± 0.28	5.23	255.3 ± 76.7
RBF fit	Endocardium	$\mathbf{0.13 \pm 0.01}$	$\mathbf{0.30}$	118.6 ± 30.2
	Epicardium	$\mathbf{0.11 \pm 0.01}$	$\mathbf{0.26}$	233.5 ± 71.4
SSM fit	Endocardium	1.82 ± 0.67	5.51	126.9 ± 32.2
	Epicardium	1.91 ± 0.76	6.23	215.2 ± 56.3

The RBF fit follows the rings almost exactly, at 0.13 mm on the endocardium and 0.11 mm on the epicardium, because it passes through the contour points by construction. The model lies 1.50 and 1.96 mm from the same rings, which is close to the in-plane pixel size of 1.37 to 1.88 mm. The SSM fit has similar mean distances, 1.82 and 1.91 mm, but a wider spread between patients: its per-patient endocardial distance ranges from 0.92 to 3.44 mm, against 1.20 to 2.14 mm for the model.

In the 30-patient ACDC pathology subset, the mean contour distance of the SSM fit is 1.42 mm for normal cases and 2.63 mm for HCM cases. The corresponding model distances are 1.45 and 1.60 mm.

Cavity volumes are similar for the model, the RBF fit and the SSM fit, at 119, 119 and 127 ml, respectively. Myocardial volumes, obtained as the epicardial minus the cavity volume, differ more, at 136, 115 and 88 ml.

Table 4.2 compares the three reconstructions with one another.

Table 4.2. Pairwise agreement between the three reconstructions of the same ED contour rings over the held-out test cohort (mean $\pm$ SD of per-patient values). Entries are symmetric disagreements, not errors. Bold marks the closest pair.

Pair	Metric	Endocardium/cavity	Epicardium/myocardium
Model vs. RBF fit	Chamfer (mm)	$\mathbf{1.57 \pm 0.18}$	$\mathbf{1.90 \pm 0.24}$
	HD95 (mm)	$\mathbf{4.24 \pm 0.86}$	$\mathbf{6.07 \pm 0.94}$
	DSC	$\mathbf{0.91 \pm 0.02}$	$\mathbf{0.77 \pm 0.05}$
Model vs. SSM fit	Chamfer (mm)	2.08 ± 0.51	2.59 ± 0.73
	HD95 (mm)	5.88 ± 1.91	6.72 ± 2.32
	DSC	0.88 ± 0.03	0.69 ± 0.04
RBF fit vs. SSM fit	Chamfer (mm)	2.01 ± 0.67	2.05 ± 0.64
	HD95 (mm)	5.88 ± 2.95	6.62 ± 2.67
	DSC	0.89 ± 0.04	0.76 ± 0.06

The model and the RBF fit form the closest pair on every metric and surface, with Chamfer distances of 1.57 mm on the endocardium and 1.90 mm on the epicardium and a myocardial DSC of 0.77. Both pairs involving the SSM fit are further apart, at 2.08 and 2.01 mm on the endocardium.

Across the three pairs, the endocardial Chamfer distance ranges from 1.6 to 2.1 mm. Cavity DSC is uniformly high, at 0.88 to 0.91, whereas myocardial DSC is lower, at 0.69 to 0.77, and falls to 0.58 for the worst patient in a comparison involving the SSM fit.

4.3.2. Slice Sensitivity

Each patient's complete ED stack is reduced to six, four and three slices. The most basal and most apical slices are always kept, so the base-to-apex extent stays the same and only the number of slices within it changes. Because the acquired stacks differ in length, a patient contributes to a given level only when its complete stack has more slices than that level. The lower rows of Figure 4.2 show the effect on one case.

Table 4.3. Deviation of each method's reduced-slice reconstruction from its complete-stack reconstruction at ED on the held-out test cohort. Bold marks the smallest deviation at each slice level.

		Chamfer (mm)		DSC		
Method	**Slices**	**Endo.**	**Epi.**	**Cavity**	**Myoc.**	ΔV_{myo} **(%)**
Proposed model	6	$\mathbf{0.79}$	0.87	$\mathbf{0.971}$	$\mathbf{0.891}$	-0.4
	4	1.27	1.48	0.953	0.819	-2.1
	3	1.83	2.17	0.934	0.782	-3.8
RBF fit	6	0.83	$\mathbf{0.83}$	0.968	0.887	$\mathbf{0.0}$
	4	$\mathbf{1.15}$	$\mathbf{0.88}$	$\mathbf{0.959}$	$\mathbf{0.858}$	$\mathbf{-1.8}$
	3	$\mathbf{1.45}$	$\mathbf{1.32}$	$\mathbf{0.947}$	$\mathbf{0.805}$	-6.2
SSM fit	6	0.89	0.95	0.968	0.883	-1.8
	4	1.46	1.39	0.942	0.812	-3.8
	3	1.94	1.92	0.921	0.775	$\mathbf{+2.6}$

Table 4.3 compares each reduced-stack reconstruction with the one the same method produces from the complete stack, so the values show the effect of removing slices rather than the differences between methods. All three methods still produce closed surfaces from three slices.

With six slices, the reconstructions change very little: surface distances stay below 1 mm and myocardial DSC is close to 0.89 for every method. With fewer slices, the RBF fit changes least, while the model and the SSM fit change by similar amounts. The difference is clearest on the epicardium, where the Chamfer distance at three slices reaches 2.17 mm for the model, 1.92 mm for the SSM fit and 1.32 mm for the RBF fit. Cavity DSC stays above 0.92 throughout, and myocardial DSC at three slices lies between 0.775 and 0.805.

Myocardial volume changes by only a few percent. At three slices, the model and the RBF fit lose 3.8% and 6.2%, whereas the SSM fit gains 2.6% after losing volume with six and four slices, so its small change at that level does not mean that it follows the complete-stack volume more closely.

4.3.3. Wall-Thickness Measurement

The four selected methods are applied to the proposed reconstruction, the RBF fit, and the SSM fit for every patient at ED. Table 4.4 reports, for each method and reconstruction, the cohort average of the per-patient mean, SD, and 5th and 95th percentiles.

Table 4.4. Wall-thickness statistics at ED over the held-out ACDC and M&Ms-2 test cohort. Each entry is the cohort average of the corresponding per-patient statistic, in millimetres.

Method	Geometry	Mean (mm)	SD (mm)	p_5 (mm)	p_{95} (mm)
Laplace field	Model	9.18	2.14	4.75	12.20
	RBF fit	8.36	2.21	5.03	12.29
	SSM fit	6.48	1.38	3.87	8.43
Yezzi–Prince	Model	10.01	1.64	7.41	12.74
	RBF fit	8.86	2.15	5.57	12.71
	SSM fit	7.10	1.23	4.75	8.88
SDF cone rays	Model	12.47	4.87	8.01	20.68
	RBF fit	9.38	2.49	5.63	13.80
	SSM fit	7.11	1.50	4.12	9.13
EDT boundary sum	Model	9.55	1.66	6.62	12.17
	RBF fit	8.51	2.00	5.42	12.05
	SSM fit	7.02	1.22	4.65	8.77

The three non-ray methods agree closely on each reconstruction, with mean thicknesses of 9.18 to 10.01 mm on the proposed reconstruction, 8.36 to 8.86 mm on the RBF fit and 6.48 to 7.10 mm on the SSM fit. The SDF cone-ray method stands apart on the first two, at 12.47 and 9.38 mm, and also spreads more widely: on the proposed reconstruction, its per-patient SD is 4.87 mm, against 1.64 to 2.14 mm for the other methods.

Table 4.5 compares patient-level mean thickness on the proposed reconstruction with the two fitting baselines.

Table 4.5. Patient-level agreement between wall thickness on the proposed reconstruction and on each fitting baseline at ED. Bias is the model value minus the baseline value. Bold marks the closest agreement in each column.

	Model vs. RBF fit			Model vs. SSM fit		
Method	**Bias (mm)**	**RMSE (mm)**	r	**Bias (mm)**	**RMSE (mm)**	r
Laplace field	$\mathbf{+0.82}$	$\mathbf{0.86}$	$\mathbf{0.99}$	$+2.70$	3.13	0.72
Yezzi–Prince	$+1.16$	1.23	$\mathbf{0.99}$	$+2.91$	3.49	$\mathbf{0.77}$
SDF cone rays	$+3.10$	3.81	0.95	$+5.36$	6.64	0.67
EDT boundary sum	$+1.03$	1.10	$\mathbf{0.99}$	$\mathbf{+2.53}$	$\mathbf{3.04}$	$\mathbf{0.77}$

With the same measurement method on each geometry, the RBF fit is thinner than the model by 0.82 mm for the Laplace field, 1.16 mm for Yezzi–Prince, 3.10 mm for SDF cone rays, and 1.03 mm for the EDT boundary sum. The SSM fit is thinner by 2.70, 2.91, 5.36, and 2.53 mm, respectively. Per-patient means are strongly correlated between the model and the RBF fit ($r = 0.99$, 0.99, 0.95, and 0.99) and less strongly between the model and the SSM fit ($r = 0.72$, 0.77, 0.67, and 0.77).

Table 4.6 compares the three other methods with the Laplace field point by point over the myocardial surface.

Table 4.6. Each wall-thickness method with the Laplace field over the myocardial surface points of the held-out test cohort. The Laplace field is the reference for comparison, not ground truth.

Method	r	MAE (mm)	RMSE (mm)	Bias (95% LoA) (mm)	ICC
Yezzi–Prince	**0.80**	0.89	2.14	$+0.83$ (± 3.86)	**0.77**
SDF cone rays	0.44	3.38	7.29	$+3.31$ (± 12.74)	0.28
EDT boundary sum	0.78	**0.81**	**2.07**	$\mathbf{+0.37}$ (± 3.99)	**0.77**

The EDT boundary sum differs from the Laplace field by 0.81 mm on average, with a bias of $+0.37$ mm, and Yezzi–Prince by 0.89 mm, with a bias of $+0.83$ mm; both reach ICC $= 0.77$, which corresponds to moderate to good agreement under the thresholds of Koo and Li [54]. The cone-ray method differs by 3.38 mm on average, with a bias of $+3.31$ mm, LoA of ± 12.74 mm and ICC $= 0.28$, which corresponds to poor agreement.

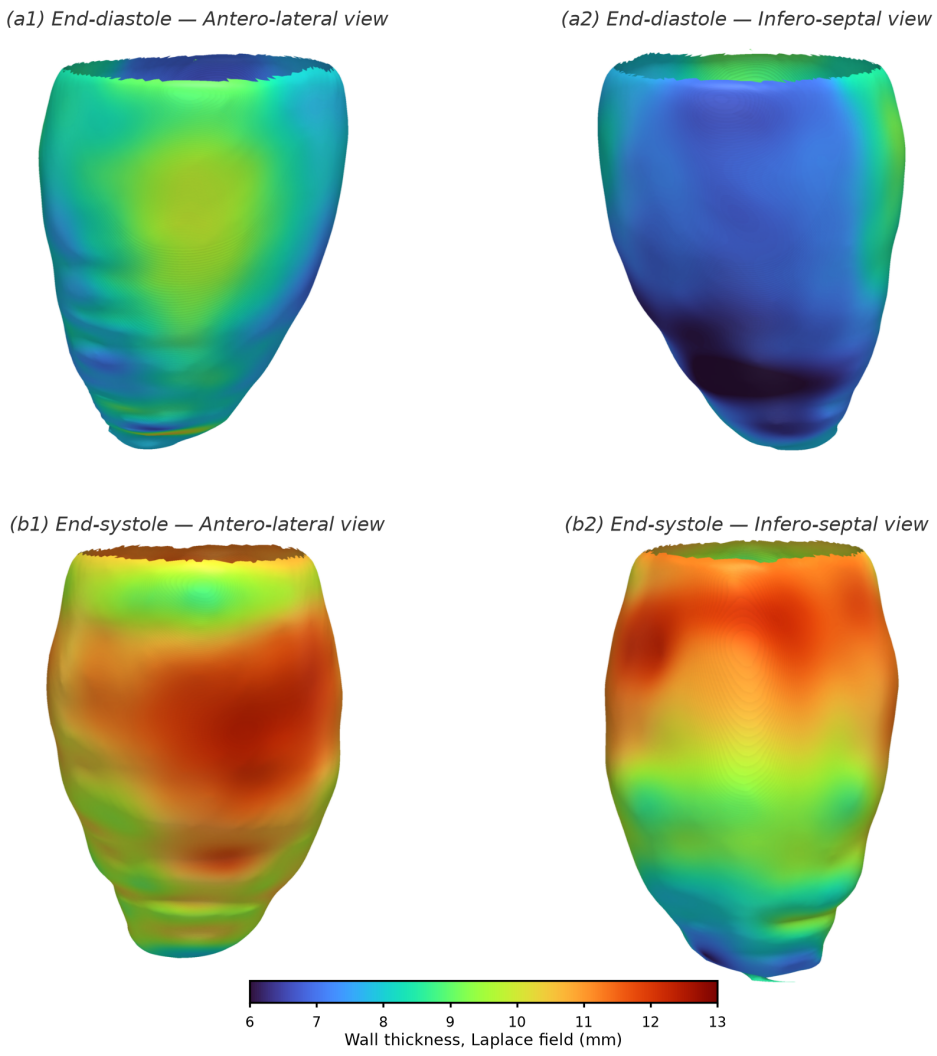


Figure 4.3. Predicted wall thickness on the reconstructed surface of one ACDC case at ED (top) and ES (bottom), shown from two viewpoints on a shared colour scale. Values are scaled to the input segmentation.

Figure 4.3 shows the predicted wall thickness as a heatmap on the reconstructed surface. The values are scaled to the input segmentation, so the map shows where the wall is thicker or thinner rather than its absolute thickness. The circumferential bands follow the spacing of the SAX slices.

4.3.4. Regional Analysis over the AHA 17-Segment Model

To localise the measurements anatomically, the reconstructed surface is divided into the 17 standard AHA segments, as described in Section 3.5.3.

The local map in Figure 4.3 resolves thickness at every surface vertex, but at that resolution it also carries the circumferential banding introduced by the slice spacing, and it cannot be compared directly across patients because no two reconstructions share the same vertices. The AHA model replaces the per-vertex values with one mean per anatomically oriented patch. Each patch spans a fixed fraction of the long axis, so its mean averages over several slice bands, and each is located by its angle relative to the septum, so segment k refers to the same region of the ventricle in every case and at both phases. The regional values therefore support the comparisons between phases and between diagnostic groups reported below, which the raw surface maps do not.

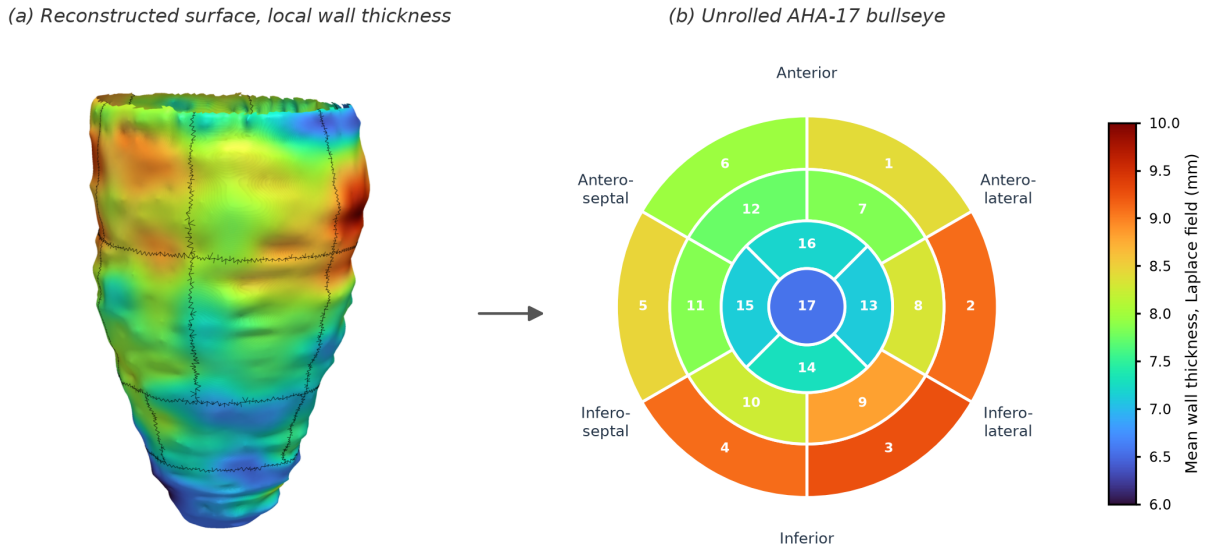


Figure 4.4. AHA 17-segment analysis of the representative case at ED: (a) reconstructed surface coloured by the mean Laplace-field wall thickness of each segment; (b) the same values on the standard bullseye, with basal (1–6), mid-cavity (7–12) and apical (13–16) rings and the apical cap (17).

Figure 4.4 shows the partition applied to the reconstructed geometry of the representative case together with a bullseye of the Laplace-derived wall thickness. Panel (a) places the result on the reconstructed surface, where the ring boundaries and the septal orientation can be checked against the geometry, while panel (b) unrolls the segment means onto the standard bullseye, with the basal ring outermost and the apical cap at the centre.

All regional values are measured with the Laplace field, for the reasons given in Section 4.3.3. Table 4.7 reports the regional pattern over the 30-patient ACDC pathology subset and its change between phases. Because the table reports cohort means, it describes the typical regional pattern of the subset rather than individual patients; the variation between patients is examined separately in Table 4.9.

Over the ACDC pathology subset the reconstructed wall averages 9.21 mm at ED and 13.87 mm at ES, corresponding to 50.6% relative thickening. At ED the septal mid-cavity segments are the thickest, while the apical cap is the thinnest at 7.76 mm. Every segment thickens between phases; the smallest relative change is 32.1% in the basal anteroseptal segment and the largest is 64.6% in the mid inferolateral segment. No fitting baseline is available at ES, so these phase values describe the model output only.

Table 4.7. Regional AHA 17-segment wall thickness on the reconstructed geometry over the 30-patient ACDC pathology subset.

#	Segment	ED (mm)	ES (mm)	Δ (%)
1	Basal anterior	9.99	13.71	$+37.1$
2	Basal anteroseptal	10.15	13.40	$+32.1$
3	Basal inferoseptal	9.89	13.43	$+35.7$
4	Basal inferior	9.09	13.81	$+51.9$
5	Basal inferolateral	8.93	14.27	$+59.8$
6	Basal anterolateral	9.53	13.87	$+45.5$
7	Mid anterior	9.19	13.79	$+50.0$
8	Mid anteroseptal	10.37	14.68	$+41.6$
9	Mid inferoseptal	10.19	15.05	$+47.6$
10	Mid inferior	9.25	14.41	$+55.8$
11	Mid inferolateral	9.09	14.96	$+64.6$
12	Mid anterolateral	8.99	14.17	$+57.6$
13	Apical anterior	8.50	13.63	$+60.3$
14	Apical septal	8.77	13.86	$+58.1$
15	Apical inferior	8.44	13.29	$+57.5$
16	Apical lateral	8.43	13.78	$+63.6$
17	Apex	7.76	11.69	$+50.7$
Overall		**9.21**	**13.87**	$\mathbf{+50.6}$

Note. Values are means for the 30 ACDC test patients used in the pathology analysis, measured with the Laplace field. Δ is the relative thickening from ED to ES.

Table 4.8 separates the normal and HCM patients, and both groups show an internally coherent regional pattern. In the normal cases the ED thickness varies little around the base and the mid-cavity, from 7.92 to 8.81 mm, and tapers towards the apex, where the apical cap at 6.80 mm is the thinnest region. Every segment thickens from ED to ES, and the thickening is graded along the long axis: the apical segments thicken by 52% to 64%, the mid-cavity segments by 51% to 64% and the basal segments by 37% to 61%. Over the normal group the mean thickness rises from 7.99 mm to 12.21 mm, a thickening of 52.9%.

Table 4.8. Regional AHA 17-segment wall thickness on the reconstructed geometry for the normal and HCM groups.

#	Segment	Normal (NOR, $n = 20$)			Hypertrophic (HCM, $n = 10$)		
		ED (mm)	ES (mm)	Δ (%)	ED (mm)	ES (mm)	Δ (%)
1	Basal anterior	8.72	12.25	$+40.4$	12.54	16.62	$+32.6$
2	Basal anteroseptal	8.81	12.10	$+37.4$	12.83	16.02	$+24.9$
3	Basal inferoseptal	8.68	11.96	$+37.8$	12.32	16.35	$+32.8$
4	Basal inferior	8.16	12.53	$+53.6$	10.96	16.37	$+49.4$
5	Basal inferolateral	7.94	12.81	$+61.3$	10.91	17.19	$+57.5$
6	Basal anterolateral	8.38	12.12	$+44.6$	11.83	17.37	$+46.9$
7	Mid anterior	7.94	12.10	$+52.5$	11.70	17.16	$+46.6$
8	Mid anteroseptal	8.55	12.98	$+51.7$	13.99	18.10	$+29.4$
9	Mid inferoseptal	8.57	12.98	$+51.4$	13.43	19.17	$+42.8$
10	Mid inferior	7.93	12.46	$+57.1$	11.89	18.32	$+54.1$
11	Mid inferolateral	7.96	13.02	$+63.5$	11.35	18.84	$+66.0$
12	Mid anterolateral	7.92	12.39	$+56.4$	11.13	17.73	$+59.3$
13	Apical anterior	7.28	11.75	$+61.4$	10.96	17.40	$+58.7$
14	Apical septal	7.49	12.00	$+60.2$	11.33	17.59	$+55.3$
15	Apical inferior	7.28	11.72	$+61.1$	10.76	16.43	$+52.6$
16	Apical lateral	7.33	12.03	$+64.0$	10.61	17.28	$+62.9$
17	Apex	6.80	10.32	$+51.8$	9.67	14.71	$+52.1$
Overall		**7.99**	**12.21**	$\mathbf{+52.9}$	**11.68**	**17.23**	$\mathbf{+47.5}$

Note. Values are group means measured with the Laplace field. Δ is the relative thickening from ED to ES. These measurements summarise the model output reconstructed from sparse SAX contours and are not an independent anatomical reference.

HCM thickness exceeds NOR thickness in every segment, with the largest differences in the mid-cavity anteroseptal and inferoseptal segments, where the HCM group reaches 13.99 and 13.43 mm compared with 8.55 and 8.57 mm in the normal group. Table 4.9 evaluates this separation at the patient level rather than treating the 17 segments as independent observations.

Table 4.9. Patient-level comparisons between the normal and HCM groups (mean $\pm$ SD). Differences are HCM minus NOR with 95% percentile-bootstrap CIs from 10 000 resamples [55]; positive Hedges' g [56] indicates a larger HCM value. Bold CIs exclude zero.

Measure	NOR	HCM	Difference (95% CI)	Hedges' g
Mean ED AHA-17 thickness (mm)	7.99 ± 0.88	11.66 ± 1.66	$\mathbf{+3.67\ [2.54,\ 4.67]}$	3.01
Maximum ED segment (mm)	9.07 ± 0.98	14.44 ± 1.94	$\mathbf{+5.37\ [4.11,\ 6.56]}$	3.82
Systolic thickening (percentage points)	53.2 ± 11.6	47.9 ± 11.5	$-5.3\ [-13.9,\ 3.0]$	-0.45

Both ED confidence intervals exclude zero, and both effect sizes are large. The interval for relative systolic thickening includes zero, so this cohort shows no clear group difference in proportional thickening. None of the normal cases and three of the ten HCM cases have a maximum segment mean above the 15 mm clinical reference line [3].

Figure 4.5 shows the patient-level distributions of absolute ED thickness and relative systolic thickening.

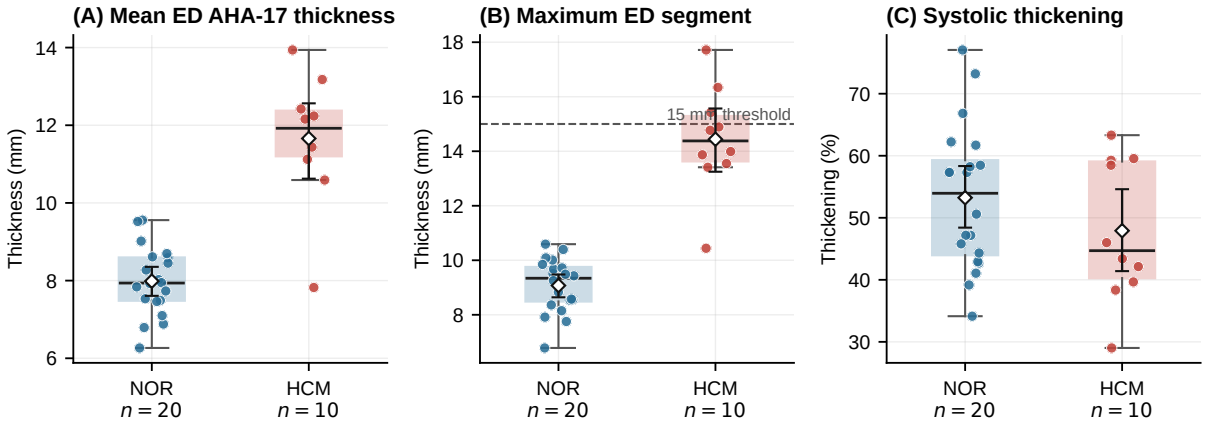


Figure 4.5. Patient-level AHA-17 wall-thickness outcomes for the normal and HCM groups. The dashed line marks the 15 mm clinical reference for maximal ED LV wall thickness; the plotted maximum segment mean is not diagnostically equivalent to it.

The two groups overlap more in relative thickening than in absolute thickness.

4.4. Discussion

The results are interpreted below against the three research questions of Section 1.3. Because no independent 3D reference is available (Section 4.1), each answer concerns fidelity to the input contours, agreement between methods, or internal consistency, not anatomical accuracy.

RQ1 compares the learned reconstruction with RBF and SSM reconstructions of the same contours. Closure does not separate the methods: every surface is watertight after the shared repair stage, so closure is a property of the pipeline rather than of the learned prior. The methods differ instead in how closely they follow the contours and in the geometry they infer between slices.

The model follows its input rings to within about one in-plane pixel (Table 4.1). Exact agreement with the rings is not the aim. Expert contours contain annotation errors and are restricted to the pixel grid, so a surface that matches them to 0.13 mm, as the RBF fit does by construction, also reproduces those errors. A consistent deviation of about one pixel is compatible with a surface that smooths this noise rather than one that misses the observations. The SSM fit has a similar mean distance but a much wider spread between patients, and the pathology subset locates the source of that spread: the SSM distance increases markedly on HCM cases, whereas the model distance changes little. A population model can represent only shapes within the variation of its training population, and thick hypertrophic walls lie partly outside it. The model, which is conditioned directly on the contours, appears less tied to an average shape.

The pairwise comparison shows where the three reconstructions disagree (Table 4.2). On the endocardium they differ by about one in-plane pixel, far less than the 10 mm slice spacing, and cavity DSC is high for every pair. Myocardial agreement is clearly lower, and myocardial volume varies by several

tens of millilitres between methods. The cause lies in what the data constrain. The contours fix the cavity outline at each slice, but both surfaces must be inferred between slices, and small changes in the inferred epicardium alter myocardial volume and thickness much more than cavity overlap. Lower myocardial agreement therefore shows that the wall geometry depends on the reconstruction prior; it does not show that one method is anatomically better. Because wall thickness is measured on this geometry, the spread between methods is a lower bound on the uncertainty of every thickness value in this chapter, and selecting one method as correct would not remove it. The model lies closest to the RBF fit, but part of this agreement is expected, because the training targets were produced by a similar surface fit (Section 4.1.2).

Slice removal degrades the three methods by similar amounts (Table 4.3). The model has the largest epicardial displacement with three slices but not the largest change in myocardial volume, so no method is more robust on every metric. These values describe stability relative to each method's own complete-stack surface, not anatomical error.

RQ2 asks whether one phase-conditioned model can reconstruct paired surfaces at ED and ES while keeping a positive wall thickness. The architecture fixes the order of the two surfaces, because the epicardial field is defined as the endocardial field minus a strictly positive offset. On the output, the Laplace-derived thickness increases in all seventeen AHA segments between ED and ES (Table 4.7), which is the expected direction of systolic wall thickening. The evaluation did not measure the minimum pointwise separation between the extracted meshes, test for surface intersections after repair, or report the ED–ES change in cavity volume, and no fitting baseline is available at ES. The evidence therefore supports a phase-dependent reconstruction with a thickening wall. Positive separation holds in the continuous fields by construction, but it has not been verified at every point of the extracted meshes, and accuracy at ES remains untested.

RQ3 asks how wall thickness depends on the reconstruction prior and on the measurement method, and whether regional and pathology-related differences are retained. Both factors change the measured thickness, in different ways.

With the method fixed, the reconstruction prior mainly shifts the absolute scale (Table 4.5). Thickness on the model and on the RBF fit is almost perfectly correlated across patients, yet the model is consistently thicker: the RBF fit preserves the ranking of patients but changes their absolute values. The SSM fit changes both the scale and, to a lesser extent, the ranking, as expected from a stronger shape prior. A high correlation therefore does not imply agreement in millimetres; only the bias reveals the offset.

With the geometry fixed, the measurement path matters (Table 4.6). Yezzi–Prince and the EDT boundary sum agree closely with the Laplace field, whereas the SDF cone-ray method is both biased and highly variable. The most likely cause is geometric. A straight ray leaves the myocardium too early or too late wherever the wall curves, whereas the Laplace and Yezzi–Prince paths follow the curvature of the wall and the EDT boundary sum does not depend on a single direction. The two effects are of similar size: the cone-ray bias of $+3.31$ mm is close to the 2.53 to 2.91 mm offset between the model and SSM geometries with the non-ray methods. Thickness values are therefore comparable only when both the reconstruction and the measurement method are the same.

For the second part of RQ3, the reconstruction retains differences between patient groups instead of reproducing an average shape. At the patient level, the HCM group is thicker at ED in both mean thickness and thickest segment, with confidence intervals that exclude zero and large effect sizes (Table 4.9). Relative systolic thickening does not separate the groups in this cohort. With ten HCM patients, these are descriptive differences within the model output rather than evidence of diagnostic ability.

4.5. Limitations of the Evaluation

The central limitation concerns the reference: no public dataset provides independently acquired 3D ground-truth surfaces for these scans, so the reconstruction is evaluated through fidelity to the observed contour rings and agreement with alternative reconstructions of those rings. The pairwise Chamfer distances measure disagreement between reconstruction priors, not distance from true anatomy. The same caveat applies to the wall-thickness comparison: the input is a stack of contours separated by several millimetres, so no method used in this chapter carries measured through-plane information, and the way that gap is inferred shifts the measured thickness. Absolute millimetre values should not be over-interpreted; the comparisons that the data support are relative ones between pipelines evaluated on identical input. The wall-thickness study compares methods and how consistently they measure thickness; it does not validate clinical use. Without expert manual wall-thickness measurements, this work can claim an objective and locally resolved computational framework, but cannot establish that it can replace clinical measurement methods.

Several further factors limit how widely these results can be applied, and the whole evaluation should be read as preliminary. The general evaluation uses held-out patients from ACDC and M&Ms-2, whereas the regional analysis by disease group is limited to 30 ACDC test patients: 20 normal and 10 with HCM. This subset is far too small to support claims of clinical or pathological generalisation, and the HCM subgroup in particular can only suggest a pattern rather than support broad conclusions. The regional analysis carries a further dependency: each patient contributes seventeen segments, so the segment values are repeated measures within a patient rather than independent samples, and they are reported as a descriptive summary for that reason. The relative thickening figures deserve particular caution, because dividing a systolic by a diastolic thickness amplifies small differences in where the wall is sampled. The regional summaries also depend on the AHA orientation, while all mesh-based measurements depend on the inference grid. A finer grid or an alternative anatomical frame could shift the values. Only the heatmap in Figure 4.3 is scaled to the input segmentation; the tables report thickness measured directly on the meshes. The reconstruction additionally inherits the quality of the input segmentation, and the training data span more than one collection, so some differences between the acquisition sources are expected. Finally, the synthetic epicardium used during pre-training is constructed rather than measured. Fine-tuning on clinical data is intended to correct for this difference, but the synthetic shapes may still influence the output. A larger cohort, and ideally a dataset with genuine 3D ground truth, would be needed to confirm the results reported here.

Reconstruction comparisons are bounded in the same way. The implicit fit and the SSM fit were chosen because their priors differ from each other and from the learned one, so the spread between the

three bounds the influence of the prior; it does not identify which is closest to the anatomy. Contour fidelity cannot settle the question either, because it evaluates only the observed rings and not the geometry between them. The slice-removal criterion in Table 4.3 may favour a stable prior that responds weakly to the observations as much as a method that generalises well from them. Because the acquired stacks vary in length, the three slice-removal levels are also supported by different subsets of patients, and the most extensive removal level rests on the longest stacks alone. The comparison does not cover spline lofting, Poisson reconstruction, or other learned models. No controlled experiment tests the effect of removing synthetic pre-training, local conditioning, phase conditioning, or positive surface coupling. The comparison describes the model alongside two conventional alternatives, but cannot show which component produced its behaviour or whether the full architecture is necessary.

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CHAPTER 5

Conclusions

This chapter closes the thesis. It summarises what was built and why, answers the three research questions of Section 1.3 on the evidence of Chapter 4, states the contributions, delimits the claims, and sets out the work that should follow.

5.1. Summary of the Work

Sparse SAX cardiac MRI presents a geometric limitation. Each slice observes the LV finely, at an in-plane pixel size below 2 mm, but the slices lie several millimetres apart, so the three-dimensional wall on which clinical measurement depends is never observed directly and has to be inferred. That inference was treated as the learning of a continuous signed-distance representation. A phase-conditioned implicit model was pre-trained on synthetic shapes sampled from an SSM and fine-tuned on ED and ES surfaces fitted to ACDC and M&Ms-2 segmentations. The model reads the sparse contour rings through a global PointNet pathway and a local feature grid, predicts the endocardial signed-distance field, and defines the epicardial field as that field minus a strictly positive, directly supervised offset, so the two surfaces cannot cross. Its output was compared with an RBF implicit fit and an SSM fit of the same rings, and myocardial wall thickness was measured on all three geometries with four local methods, summarised across the AHA 17 segments, and compared between normal and HCM patients.

5.2. Answers to the Research Questions

RQ1 asked how a reconstruction learned from synthetic and clinical geometry compares with RBF and SSM reconstructions of the same sparse contours. The model follows its input contours to within about one in-plane pixel (1.50 mm on the endocardium and 1.96 mm on the epicardium), lies closer to the RBF fit than to the SSM fit, and, unlike the SSM fit, keeps a similar contour distance on normal and HCM cases. The three methods agree on the cavity but not on the wall, and removing slices degrades all three by similar amounts. The learned reconstruction therefore performs comparably to conventional fitting and handles abnormal anatomy more evenly than the population model, but it is not more robust to slice removal.

RQ2 asked whether a single phase-conditioned model can reconstruct paired endocardial and epicardial surfaces at ED and ES while maintaining a positive wall thickness. The answer is a qualified yes. One network reconstructs both phases, the epicardial field cannot cross the endocardial field by construction, and the reconstructed wall thickens in every AHA segment, from 9.21 mm at ED to 13.87 mm at ES. Pointwise separation on the extracted meshes and accuracy at ES were not verified.

RQ3 asked how strongly wall thickness depends on the reconstruction prior and the measurement method, and whether regional and pathology-related differences are retained. Both factors matter. The

prior mainly shifts the absolute scale while largely preserving the ranking of patients, and a straight-ray measurement departs strongly from methods that follow the curved wall. Absolute thickness values are therefore comparable only within the same reconstruction and measurement pipeline. The reconstructed geometry does retain group differences: the HCM group is 3.67 mm thicker than the normal group in patient-level mean ED thickness, whereas relative systolic thickening does not separate the groups. With ten HCM patients, this is a descriptive result rather than a diagnostic one.

5.3. Contributions

This work makes three contributions. First, a shared data pipeline combines synthetic SSM meshes with clinical ED and ES segmentations from two public datasets. It stores contour rings, surface samples, signed-distance targets, and wall-offset targets in one format, allowing the model to be pre-trained on population shapes and fine-tuned on patient data without changing how the data are represented. Second, the phase-conditioned model processes endocardial and epicardial contour information separately, combines a global shape summary with local contour information, and uses a supervised positive offset to keep the epicardial field outside the endocardial field. Third, the evaluation approach assesses how closely reconstructions follow the observed rings, how much reconstructions from different methods agree, and how results change when slices are removed. It also compares wall-thickness methods on the same geometry, then summarises thickness by AHA segment and compares patient groups. This evaluation approach can be applied to other networks.

5.4. Limitations

The limitations discussed in Section 4.5 bound all of these answers. The most important is the absence of an independent 3D reference: the geometric results measure fidelity to the contours and agreement between methods, and the thickness results measure reproducibility rather than accuracy. The pathology cohort is small, and no ablation experiment isolates the contribution of individual model components.

5.5. Future Work

These limitations set the order of the next steps. The first need is an anatomical reference. Higher-resolution volumetric acquisitions, or expert three-dimensional surface and wall-thickness annotations on a larger external cohort, would turn the agreement figures reported here into accuracy figures and would show which of the three priors lies closest to the true wall. The second is a wider and fully matched comparison, adding spline lofting, Poisson reconstruction and learned alternatives under the same patients, physical spacing, repair procedure and metrics, with error maps showing where reconstructions disagree at the apex, the base and the most curved parts of the wall. The third is a set of controlled tests that remove synthetic pre-training, local conditioning, phase conditioning and the positive-offset coupling one at a time to show which components drive the model's behaviour. Adding pointwise separation and intersection checks on the extracted meshes would complete the answer to RQ2. Replacing the binary phase label with a continuous position in the cardiac cycle would then allow the representation to be tested for coherence in time as well as continuity in space.

In sum, this thesis shows that a phase-conditioned implicit representation can turn a sparse SAX stack into closed, paired LV surfaces that honour the observed contours, thicken consistently between diastole and systole, and support locally resolved wall-thickness analysis. It also shows, through the disagreement between equally plausible reconstructions of the same rings, how much of the myocardial wall is still inferred rather than observed, and therefore how carefully any thickness derived from it must be read.

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CHAPTER A

Supplementary Methodological Details

This appendix summarises the training data streams and partitions, RBF-derived clinical supervision meshes, training-loss terms, and model configuration described in Chapter 3.

A.1. Training Data Streams

Synthetic meshes establish a geometric shape prior during pre-training; clinical MRI data support fine-tuning on end-diastolic and end-systolic anatomy.

Table A.1. Training data streams and their roles.

Stream	Phase	Source	Training role
Synthetic SSM	ED	UK Digital Heart Project LV SSM	Pre-training to establish a geometric shape prior
Real MRI	ED	ACDC and M&Ms-2	Fine-tuning on patient-specific diastolic anatomy
Real MRI	ES	ACDC and M&Ms-2	Fine-tuning on systolic anatomy with phase conditioning

A.2. Dataset Composition and Partitions

Clinical samples are divided into training, validation, and test partitions, while accepted synthetic meshes are reserved for pre-training.

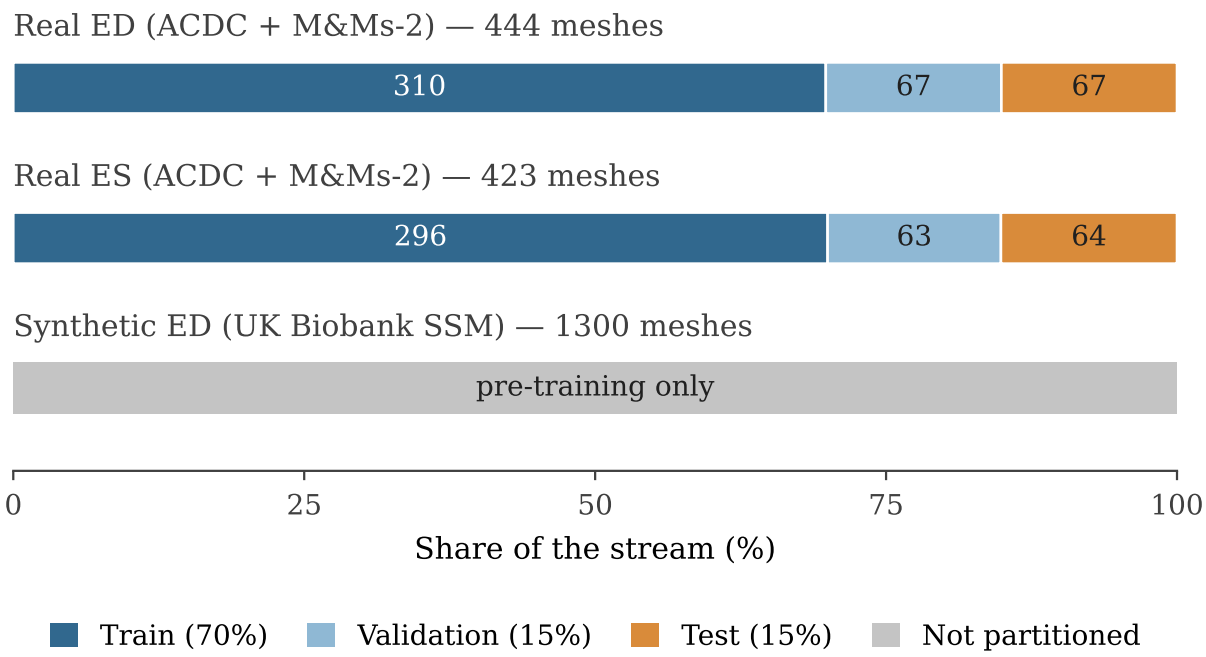


Figure A.1. Valid clinical samples divided into the 70 %/15 %/15 % partitions. The 1300 accepted synthetic meshes are reserved for pre-training.

A.3. Clinical Mesh Preparation

The clinical supervision meshes are constructed from SAX contours using RBF interpolation, surface extraction, and geometric validation.

Table A.2. Operations used to construct the RBF-derived clinical supervision meshes.

Operation	Purpose
RAS+ canonicalisation	Standardises patient-coordinate orientation while preserving physical spacing
Label harmonisation	Maps ACDC and M&Ms-2 labels to common endocardial and epicardial definitions
Largest-ring extraction and resampling	Retains 60 ordered points per boundary on each acquired SAX slice
In-plane normal estimation	Defines an outward direction at every contour point
Off-surface constraints at ± 2.5 mm	Establishes the sign and local scale of each implicit field
Thin-plate-spline RBF fit	Interpolates a continuous field with degree 1 and regularisation parameter 0.5
Evaluation on a 1.0 mm grid	Discretises the continuous field for surface extraction
Signed-distance regularisation ($\sigma = 1.2$ mm)	Reduces voxel-scale corrugation before surface extraction
Connected-region filtering and marching cubes	Removes invalid regions and extracts the zero level set
Topology validation and conditional correction	Preserves valid RBF surfaces and corrects only failed shells
Geometric plausibility checks	Rejects failed fits and invalid endocardial–epicardial pairs

A.4. Training-Loss Terms

The five loss terms combine surface and volumetric supervision with signed-distance regularisation and constraints on the decoder wall offset.

Table A.3. Purpose of the five training-loss terms.

Term	Purpose	Effect on reconstruction
$\mathcal{L}_{\text{surf}}$	Fits zero-level sets to surface samples	Places endocardial and epicardial surfaces on the target meshes
$\mathcal{L}_{\text{SDF}}$	Fits cached query-point distances	Provides dense volumetric supervision
$\mathcal{L}_{\text{eik}}$	Encourages $\|\nabla f\| = 1$	Regularises the decoder as a signed-distance field
$\mathcal{L}_{\text{off}}$	Repels off-surface points from zero	Reduces spurious zero crossings away from the myocardium
$\mathcal{L}_{\text{WT}}$	Fits the predicted wall offset to wall targets	Constrains separation of the predicted fields

A.5. Model Configuration

The configuration below specifies the contour conditioning, decoder architecture, output fields, and inference grid used by the model.

Table A.4. Main architecture settings of the model.

Component	Configuration
Input features	3D coordinates, tissue label, and cardiac phase
Global encoder type	Shared network with endocardial and epicardial max-pooling
Local conditioning	Coarse 3D contour feature volume with trilinear interpolation
Latent dimension	256
Local volume size	16^3 with 32 channels
Positional encoding	Fourier features with $L = 3$ frequency bands
Decoder	8-layer network, width 512, skip connection at layer 4
Activation	Softplus (smooth)
Local injections	Local feature paths into the decoder trunk and offset head
Output heads	Endocardial signed distance and positive decoder wall-offset field
Offset mapping	Sigmoid component with $\tau_{\text{min}} = 0.05$ and $\delta_{\text{cap}} = 0.45$, plus learned soft-hinge headroom
Inference grid	96^3 query grid for marching cubes

A.6. Training Configuration

The settings below specify the optimisation, schedule, batching, and hardware used to train the model in both stages.

Table A.5. Training configuration used for the model.

Setting	Value
Training strategy	SSM pre-training, then fine-tuning on real ED/ES
Training mode	Combined ED and ES with phase conditioning
Augmentation	Contour-input perturbations for synthetic ED and real ED/ES; targets unchanged
Optimiser	AdamW
Learning rate	2×10^{-5}
Scheduler	Cosine annealing with warm restarts ($T_0 = 100$ epochs)
Weight decay	5×10^{-4}
Training duration	Approximately 200 pre-training epochs, then 776 fine-tuning epochs (best checkpoint: 695)
Batch size	8 per GPU $\times$ 2 GPUs, accumulated for 2 steps (effective batch size 32)
Gradient clipping	1.0
Precision	Automatic mixed precision
Hardware	2 GPUs in parallel
Inference grid	96^3